Math 50 Notes: Estimators, Uncertainty, and Regression

Lectures 8–13 (10/2–10/14) and the Unit 3 course notes

How to use these notes

These notes combine the slides from 10/2, 10/5, 10/7, 10/9, 10/12 and 10/14 with the Unit 3 course notes, organized by topic instead of by lecture. Every derivation is written out step by step, with the reason for each step in gray on the right.

New to the notation? Start with the Glossary right after the contents: every symbol, how to read it aloud, what it means, and an example. Then read Section 1, which reviews the Unit 1–2 rules (from Lectures 1–7) with a small worked example for each. If a step later confuses you, the rule it needs is in Section 1.

Green boxes are worked examples (most of them are the in-class questions, with solutions). Blue boxes are the results to remember; each ends with an *In words* line that reads the formula in plain English. The last page is a one-page formula sheet with a short symbol key.

The figures are *illustrations* of the pictures in the slides, drawn from formulas or from made-up data; they are not the real datasets. All code is Python, exactly as in the slides and notes.

Contents

Glossary of notation **4**

1 Toolkit: the probability and algebra used everywhere **8**

1.1 Probability notation from Unit 1 . . . 8

1.2 Sums (Σ notation) . . . 9

1.3 Expectation and variance . . . 9

1.4 Conditioning: what $Y \mid X$ and "conditional on X" mean . . . 10

1.5 Covariance and correlation . . . 11

1.6 Normal, Bernoulli, Binomial . . . 11

1.7 The single-predictor regression model (from Unit 2) . . . 12

2 Correlation and the coefficient of determination (Unit 2.4) **12**

2.1 Correlation is the slope between standardized variables . . . 12

2.2 The coefficient of determination ρ^2 . . . 13

3 The Law of Large Numbers and the Central Limit Theorem (Unit 3.1) **14**

4 Estimators and the sample distribution (Unit 3.2) 15

4.1 Estimators . . . 15

4.2 Replicates and the sample distribution . . . 15

5 Bias, consistency, and the bias–variance trade-off (Unit 3.2) 17

5.1 Definitions . . . 17

5.2 Laplace's rule of succession . . . 17

5.3 Estimating the variance: Bessel's correction . . . 19

6 Confidence intervals (Unit 3.3) 20

6.1 True vs. estimated sample distribution . . . 20

6.2 The 95% confidence interval . . . 20

6.3 Interpreting a CI correctly (optional subtlety) . . . 21

7 Hypothesis testing (Unit 3.6) 21

7.1 p-values are not R^2 . . . 23

8 Least squares for single-predictor regression (Unit 3.4) 24

8.1 The estimators . . . 24

8.2 Residuals, SSR, SST, $\hat{\sigma}_\epsilon^2$, R^2 . . . 24

8.3 Why "least squares" . . . 25

8.4 Two kinds of replicates . . . 25

8.5 $\hat{\beta}_1$ is unbiased . . . 26

8.6 The sample distribution of $\hat{\beta}_1$ . . . 26

8.7 Computing everything in Python . . . 27

9 Autoregressive models and the Hurwicz bias (Unit 3.5) 28

9.1 Time series and the AR(1) model . . . 28

9.2 Stationary AR(1) processes . . . 28

9.3 When $\rho \approx 1$: the random walk . . . 29

9.4 Why least squares is biased here: the Hurwicz bias . . . 29

9.5 How fast do correlations decay? Autocorrelation . . . 31

9.6 Forecasting: random walk vs. linear extrapolation . . . 31

9.7 Prices vs. log prices (optional) . . . 32

10 Study design and multiple predictors (Unit 4.1–4.2) 32

10.1 Two ways to collect data . . . 32

10.2 Example: children's test scores . . . 33

10.3 What we computed vs. what we want 33
10.4 From two predictors to one 33
10.5 Which predictor matters more? 34
10.6 R^2 with multiple predictors 35

11 Omitted-variable bias and Simpson's paradox 36
11.1 The theorem 36
11.2 Simpson's paradox 37

Glossary of notation

Read the "Read as" column out loud: most of the confusion with math notation goes away once you can *say* a formula. Entries are grouped by topic; within each group the symbols appear roughly in the order you will meet them.

Symbol	Read as	Meaning	Example
A. General math			
$=$, $\approx$	"equals", "is approximately"	exactly equal vs. close to	$\sqrt{42} \approx 6.48$
$\propto$	"is proportional to"	changes by the same factor	CI width $\propto 1/\sqrt{N}$: $4\times$ the data, half the width
$\Longrightarrow$	"implies", "so"	the left side forces the right side	$\sqrt{N} \geq 7.84 \Longrightarrow N \geq 61.5$
$\Longleftrightarrow$	"if and only if"	the two statements are equivalent (each implies the other)	$\lvert\hat{T}\rvert > 1.96 \iff$ the CI excludes 0
$\to$	"tends to", "approaches"	gets closer and closer to	$\text{se}(\overline{Y}) \to 0$ as $N \to \infty$
∞	"infinity"	without limit; $N \to \infty$ means "as we collect more and more data"	
$\geq$, $\leq$, $\gtrsim$	"at least", "at most", "roughly at least"	comparisons	$N \gtrsim 20$: about 20 or more
$\sum_{i=1}^{N} a_i$	"the sum from i equals 1 to N of a sub i"	$a_1 + a_2 + \cdots + a_N$. Plain $\sum_i$ means "over all data points"	$\sum_{i=1}^{3} i = 1+2+3 = 6$
Y_i	"Y sub i", "Y i"	the i-th data point	Y_3 is the third observation
$X_{k,i}$	"X sub k comma i"	value of predictor k for data point i	$X_{2,5}$: the IQ of mother 5, if predictor 2 is IQ
Y_t, Y_{t-1}	"Y sub t", "Y sub t minus 1"	value at time t; value one step earlier	today's and yesterday's log price
$\ldots$, $\cdots$	"and so on"	continue the obvious pattern	$Y_1, \ldots, Y_N$ = all N data points
$\lvert x\rvert$	"absolute value of x"	size of x, ignoring its sign	$\lvert -0.4\rvert = 0.4$
$\sqrt{x}$, x^2	"square root of x", "x squared"		$\sqrt{3/2} \approx 1.22$
e^x, $\ln x$, $\log_{10} x$	"e to the x", "natural log", "log base 10"	exponential function and its inverses	$e^x \approx 1 + x$ for small x
$\int_a^b f(y)\,dy$	"integral from a to b of f"	area under the curve f between a and b	$\mathbb{P}(a < Y < b) = \int_a^b f(y)\,dy$ (Lecture 4)
$\in$	"is in", "belongs to"	membership in a set or interval	$y \in S$; $\rho \in [-1, 1]$
$\{\ldots\}$	"the set"	a list of possible values	a die: $S = \{1, 2, 3, 4, 5, 6\}$
$[a, b]$, (a, b)	"closed / open interval from a to b"	all numbers between a and b, with / without the endpoints	a probability is in $[0, 1]$

Symbol	Read as	Meaning	Example
$\operatorname{argmin}_b f(b)$	"the b that minimizes f"	the *input* that gives the smallest output (not the smallest value itself)	least squares: $(\hat{\beta}_0, \hat{\beta}_1)$ is the argmin of SSR
$\square$	"end of proof"	the argument is finished	
$'$ (prime)	"prime"	**two meanings**: $F'(y)$ is a derivative (Lecture 4); β' is a single-predictor coefficient (Lectures 12–13)	$\beta'_{hs} = 11.8$ vs. $\beta_{hs} = 5.95$

B. Probability (Units 1–2)

Y vs. y	"capital Y", "little y"	capital = random variable (not yet known); lowercase = one particular value it could take	$\mathbb{P}(Y = y)$: chance Y comes out as y
S	"the sample space"	every possible outcome	a coin: $S = \{0, 1\}$
E, $\mathbb{P}(E)$	"event E", "probability of E"	an event is a set of outcomes	$\mathbb{P}(Y \text{ even}) = 3/6 = 1/2$ for a die
$P_Y(y)$	"P sub Y of y"	the distribution of Y: the probability of each outcome, $\mathbb{P}(Y = y)$	fair die: $P_Y(y) = 1/6$
$\mathbb{P}(x, y)$	"probability of x and y"	**joint**: both happen	$\mathbb{P}(C{=}1, G{=}M) = 0.0021$
$\mathbb{P}(x \mid y)$	"probability of x given y"	**conditional**: among the cases with y, the fraction with x	$\mathbb{P}(\text{sad} \mid \text{brown}) = 4/16 = 1/4$
$Y \mid X$	"Y given X"	the distribution of Y once X is known	$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$
$\sim$	"is distributed as", "follows"	names the distribution of a random variable	$Y \sim \text{Bernoulli}(0.75)$: a free throw
iid, $\overset{\text{iid}}{\sim}$	"independent and identically distributed"	separate draws from the same distribution that do not affect each other	N coin flips
Bernoulli(q)	"Bernoulli q"	0 or 1, with $\mathbb{P}(1) = q$	made free throw, $q = 0.75$
Binomial(N, q)	"Binomial N, q"	number of 1's in N iid Bernoulli(q)	heads in 200 flips
Uniform(a, b)	"Uniform from a to b"	equally likely to land anywhere in $[a, b]$	bus wait $T \sim \text{Uniform}(0, 5)$
Normal(μ, σ^2)	"Normal with mean μ, variance σ^2"	the bell curve. The second number is the *variance*, not the SD	Normal(0, 1): the standard Normal
$f(y)$	"f of y", "the density"	height of the curve; probabilities are *areas* under it	
$F(y)$	"F of y", "the CDF"	$\mathbb{P}(Y \le y)$: area to the left of y	
Z, $\Phi(z)$	"Z", "capital phi of z"	$Z \sim \text{Normal}(0, 1)$; $\Phi(z) = \mathbb{P}(Z < z)$	$\Phi(1.96) \approx 0.975$
$\mathbf{1}_A$	"indicator of A"	1 if A happens, 0 if not	$\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$; `np.mean(x==2)`
$\mathbb{E}[Y]$	"expected value of Y", "E of Y"	the long-run average, $\sum_y y\, P_Y(y)$	a fair die: 3.5

Symbol	Read as	Meaning	Example
$\mathbb{E}[Y \mid X]$	"expected Y given X"	average of Y within the group with that value of X	$\mathbb{E}[Y \mid X{=}1] - \mathbb{E}[Y \mid X{=}0] = \beta_1$
$\mathrm{var}(Y)$, σ^2	"variance of Y", "sigma squared"	average squared distance from the mean	Bernoulli(q): $q(1-q)$
σ, σ_Y	"sigma (sub Y)"	standard deviation = $\sqrt{\text{variance}}$; same units as Y	test scores: $\sigma_Y = 20.4$
$\mathrm{cov}(X, Y)$	"covariance of X and Y"	do X and Y move together? + together, − opposite, 0 no linear link	$\mathrm{cov}(X, Y) = \beta_1 \sigma_X^2$
ρ, $\mathrm{corr}(X, Y)$	"rho", "correlation"	covariance rescaled to lie in $[-1, 1]$	$\hat{\rho} \approx 0.45$ (mom's IQ, child's score)
ρ_k	"rho sub k"	autocorrelation: correlation of values k steps apart	AR(1): $\rho_k = \rho^k$

C. Greek letters

μ	"mew" (mu)	a mean (true average)	$\mathbb{E}[Y] = \mu$
σ, Σ	"sigma", "capital sigma"	σ: standard deviation; Σ: the sum sign	
ρ	"row" (rho)	correlation	
β	"bay-tuh" (beta)	regression coefficients	β_0 intercept, β_1 slope
ϵ	"epsilon"	noise: the part of Y the line does not explain	$Y = \beta_0 + \beta_1 X + \epsilon$
θ	"thay-tuh" (theta)	a generic parameter, when the specific one does not matter	$\hat{\theta}$ estimates θ
Φ	"fye" (capital phi)	the standard Normal CDF	
π	"pie"	the number 3.14159…; appears in the Normal density	
λ	"lamb-duh" (lambda)	a rate parameter (Lecture 4 only)	

D. Data, estimators, and regression

N, K	"N", "K"	number of data points (or length of a time series); number of predictors	$N = 434$ children, $K = 2$
$\hat{\theta}$ (hat)	"theta hat"	an **estimate** computed from data (random) of the true θ (fixed)	$\hat{q} = 3/12$ estimates q
$\overline{Y}$ (bar)	"Y bar"	sample average $\frac{1}{N}\sum_i Y_i$	$\overline{Y}$ of 2, 4, 6 is 4
$\mathrm{se}(\hat{\theta})$	"standard error of theta hat"	SD of the estimate over replicates of the data	$\mathrm{se}(\overline{Y}) = \sigma/\sqrt{N}$
MSE, Bias	"mean squared error", "bias"	average squared miss; average miss $\mathbb{E}[\hat{\theta}] - \theta$	$\mathrm{MSE} = \mathrm{se}^2 + \mathrm{Bias}^2$
$\hat{\sigma}^2$, $\hat{\sigma}_0^2$	"sigma hat squared (sub zero)"	variance estimates, dividing by $N-1$ / by N	Bessel's correction
S, $\hat{q}_L$	"S", "q hat sub L"	number of 1's, $\sum_i Y_i$; Laplace's estimate $\frac{S+1}{N+2}$	two tails: $\hat{q}_L = 1/4$, not 0
CI	"confidence interval"	$\hat{\theta} \pm 1.96\ \mathrm{se}(\hat{\theta})$	graders: $[-2, 3]$

Symbol	Read as	Meaning	Example
H_0	"H naught", "the null hypothesis"	the claim we test, e.g. "no effect"	$H_0 : \beta_1 = 0$
$\hat{T}$, $\hat{T}^*$	"T hat", "T star"	test statistic from our data / from a replicate where H_0 is true	graders: $\hat{T} \approx 0.4$
p_v	"p-value"	$\mathbb{P}(\lvert\hat{T}^*\rvert > \lvert\hat{T}\rvert)$: how surprising our data would be if H_0 were true	graders: 0.69
$\hat{Y}_i$, r_i	"Y hat i", "residual i"	fitted value $\hat{\beta}_0 + \hat{\beta}_1 X_i$; miss $Y_i - \hat{Y}_i$	
SSR, SST	"sum of squared residuals", "total sum of squares"	$\sum_i r_i^2$; $\sum_i (Y_i - \overline{Y})^2$	
S_{xx}	"S x x"	$\sum_i (X_i - \overline{X})^2 \approx N\sigma_X^2$	
R^2	"R squared"	$1 - \text{SSR/SST}$: fraction of the spread in Y explained; estimates ρ^2	test scores: 0.21
β_0, β_1	"beta zero", "beta one"	intercept and slope of the regression line	
β_{hs}, β_{iq}	"beta H S", "beta I Q"	coefficient on the named predictor	$\hat{\beta}_{iq} = 0.56$
β'	"beta prime"	coefficient when that predictor is *alone* in the model	$\hat{\beta}'_{hs} = 11.8$
$\beta_{1,2}$	"beta one two"	slope when X_2 is regressed on X_1 (the *second* on the *first*)	$\beta_{hs,iq} = 10.3$: IQ gap between the HS groups
σ_ϵ	"sigma epsilon"	SD of the noise: spread of points around the line	$\hat{\sigma}_\epsilon \approx 5$ (graders)

E. Python

`np.mean(x)`, `np.sum(x)`, `len(x)`	average, total, count	`np.mean(x)` = `np.sum(x)/len(x)`
`np.var(x)`, `np.std(x)`	variance and SD (dividing by N)	
`np.cov(x,y)[0,1]`	sample covariance (dividing by $N-1$); `[row, column]` counted from 0	$\hat{\beta}_1$ = `C[0,1]/C[0,0]`
`y[x==0]`	the y values where x is 0: data for "$Y \mid X = 0$"	`np.mean(y[x==0])`
`x**2`	x squared, element by element	
`Y[:-1]`, `Y[1:]`	all but the last; all but the first	yesterday/today pairs
`np.random.binomial(N,q)`	one Binomial(N, q) draw	
`rng.normal(size=N)`	N draws from Normal$(0, 1)$	
`norm.cdf(z)`	$\Phi(z)$ (from `scipy.stats`)	`norm.cdf(1.96)` ≈ 0.975
`sm.OLS(y, X).fit()`	least squares fit with statsmodels	Section 10

1 Toolkit: the probability and algebra used everywhere

Almost every derivation in these notes uses only the facts below. They come from Units 1–2 (Lectures 1–7); each has a small worked example from those lectures.

1.1 Probability notation from Unit 1

A **random variable** Y is a quantity determined by chance (capital letter). It takes values in a **sample space** S; the elements of S are **outcomes**, and sets of outcomes are **events**. The **distribution** $P_Y(y) = \mathbb{P}(Y = y)$ gives the probability of each outcome; these are ≥ 0 and add up to 1. We name distributions with $\sim$: $Y \sim \text{Bernoulli}(q)$ means $S = \{0, 1\}$ and $P_Y(1) = q$.

For two variables we use three kinds of probability:

- **Joint** $\mathbb{P}(x, y)$: both happen.
- **Marginal** $\mathbb{P}(x) = \sum_y \mathbb{P}(x, y)$: "sum out" the variable you do not care about.
- **Conditional** $\mathbb{P}(x \mid y) = \dfrac{\mathbb{P}(x, y)}{\mathbb{P}(y)}$: among the cases where y happened, the fraction where x also happened.

X and Y are **independent** if $\mathbb{P}(x, y) = \mathbb{P}(x)\mathbb{P}(y)$ for *every* pair (x, y); equivalently, knowing one does not change the probabilities of the other.

Examples (Lectures 1–2)

A die. $S = \{1, \ldots, 6\}$, $P_Y(y) = 1/6$. The event "Y even" is $\{2, 4, 6\}$, so $\mathbb{P}(Y \text{ even}) = 3/6 = 1/2$.
Cancer diagnosis (C) and gender (G). The joint table:

$\mathbb{P}(C, G)$	$G = M$	$G = F$	sum
$C = 1$ (diagnosed)	0.0021	0.0019	0.0040
$C = 0$	0.4899	0.5061	0.9960
sum	0.4920	0.5080	1

Marginal: $\mathbb{P}(C = 1) = 0.0021 + 0.0019 = 0.004$ (sum across the row). *Conditional:* $\mathbb{P}(C = 1 \mid G = M) = 0.0021/0.492 = 0.00427$ (restrict to the M column, then divide by its total). *Independent?* Compare one cell with the product of its margins: $\mathbb{P}(C{=}1)\mathbb{P}(G{=}M) = 0.004 \times 0.492 = 0.00197 \neq 0.0021$, so not exactly.
Eye color and mood. Of 24 people, 16 have brown eyes and 4 of those are sad, so $\mathbb{P}(\text{sad} \mid \text{brown}) = 4/16 = 1/4$, while the joint probability is $\mathbb{P}(\text{sad}, \text{brown}) = 4/24 = 1/6$. Conditioning shrinks the denominator to the group you condition on.

1.2 Sums (Σ notation)

$\sum_{i=1}^{N} a_i$ means $a_1 + a_2 + \cdots + a_N$. We often write just $\sum_i$. Three rules:

$$\sum_i c\,a_i = c\sum_i a_i \qquad \text{(a constant } c \text{ comes out of the sum)}$$

$$\sum_i (a_i + b_i) = \sum_i a_i + \sum_i b_i \qquad \text{(a sum splits)}$$

$$\sum_{i=1}^{N} c = Nc \qquad \text{(adding the same number } N \text{ times)}$$

The **sample average** is $\overline{Y} = \frac{1}{N}\sum_i Y_i$, so $\sum_i Y_i = N\overline{Y}$. An important consequence: deviations from the sample average always sum to zero,

$$\sum_i (Y_i - \overline{Y}) = \sum_i Y_i - N\overline{Y} = N\overline{Y} - N\overline{Y} = 0.$$

Because of this, for any constant c (including $c = \overline{X}$ or $c = \overline{Y}$), $\sum_i c\,(Y_i - \overline{Y}) = c \cdot 0 = 0$. We will use this trick several times.

Example

Data $Y_1, Y_2, Y_3 = 2, 4, 6$. Then $\sum_i Y_i = 12$, $\overline{Y} = 12/3 = 4$, and the deviations $Y_i - \overline{Y}$ are $-2, 0, 2$, which sum to 0. Rule 1: $\sum_i 5Y_i = 10 + 20 + 30 = 60 = 5 \times 12$. Rule 3: $\sum_{i=1}^{3} 7 = 21 = 3 \times 7$. In Python (Lecture 3), `np.mean(x)` is exactly `np.sum(x) / len(x)`.

1.3 Expectation and variance

The **expectation** (true mean) is the probability-weighted average of the outcomes, $\mathbb{E}[Y] = \sum_y y\,P_Y(y)$ (for a density, $\mathbb{E}[Y] = \int y\,f(y)\,dy$), and more generally $\mathbb{E}[h(Y)] = \sum_y h(y)P_Y(y)$. It is what the sample average approaches as N grows: in an iid sample, the fraction of times outcome y appears approaches $P_Y(y)$, with an error proportional to $1/\sqrt{N}$ (Lecture 3; Section 3 makes this precise).

- **Linearity:** $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$ for any random variables (independent or not). $\mathbb{E}[c] = c$ for a constant.
- **Indicators:** a probability is the expectation of a 0/1 variable: $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. That is why `np.mean(x == 2)` estimates $\mathbb{P}(X = 2)$.
- **Variance:** $\operatorname{var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2$, and $\sigma_X = \sqrt{\operatorname{var} X}$.
- **Shifting and scaling:** $\operatorname{var}(aX + b) = a^2 \operatorname{var}(X)$. Adding a constant does not change the spread; multiplying by a multiplies the spread by $|a|$ and the variance by a^2.
- **Independent sums:** if X, Y are independent, $\operatorname{var}(X+Y) = \operatorname{var} X + \operatorname{var} Y$ and $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. Note $\operatorname{var}(X - Y) = \operatorname{var} X + \operatorname{var} Y$ too (since $\operatorname{var}(-Y) = (-1)^2 \operatorname{var} Y$).

Examples (Lectures 5–6)

A three-valued variable. $\mathbb{P}(Y{=}1) = \frac{1}{2}$, $\mathbb{P}(Y{=}2) = \frac{1}{3}$, $\mathbb{P}(Y{=}3) = \frac{1}{6}$. Then

$$\mathbb{E}[Y] = 1{\cdot}\tfrac{1}{2}{+}2{\cdot}\tfrac{1}{3}{+}3{\cdot}\tfrac{1}{6} = \tfrac{5}{3}, \qquad \mathbb{E}[Y^2] = 1{\cdot}\tfrac{1}{2}{+}4{\cdot}\tfrac{1}{3}{+}9{\cdot}\tfrac{1}{6} = \tfrac{10}{3}, \qquad \mathrm{var}(Y) = \tfrac{10}{3} - \left(\tfrac{5}{3}\right)^2 = \tfrac{30}{9} - \tfrac{25}{9} = \tfrac{5}{9}.$$

Note $\mathbb{E}[Y^2]$ means "square first, then average" ($\frac{10}{3}$), which is not $(\mathbb{E}Y)^2$ ($\frac{25}{9}$).
Bernoulli variance. $Y \sim \text{Bernoulli}(q)$: $\mathbb{E}[Y] = 0 \cdot (1-q) + 1 \cdot q = q$. Since Y is 0 or 1, $Y^2 = Y$, so $\mathbb{E}[Y^2] = q$ and $\mathrm{var}(Y) = q - q^2 = q(1-q)$. It is largest ($\frac{1}{4}$) at $q = \frac{1}{2}$ and zero at $q = 0$ or 1 (no randomness).

The variance of a sample average

If $Y_1, \ldots, Y_N$ are iid (independent, identically distributed) with mean μ and variance σ^2, then

$$\mathbb{E}[\overline{Y}] = \tfrac{1}{N}\textstyle\sum_i \mathbb{E}[Y_i] = \tfrac{1}{N} \cdot N\mu = \mu,$$
$$\mathrm{var}(\overline{Y}) = \mathrm{var}\Big(\tfrac{1}{N}\textstyle\sum_i Y_i\Big) = \tfrac{1}{N^2}\,\mathrm{var}\Big(\textstyle\sum_i Y_i\Big) = \tfrac{1}{N^2} \cdot N\sigma^2 = \frac{\sigma^2}{N}.$$

The step $\mathrm{var}(\sum_i Y_i) = N\sigma^2$ uses independence. So the standard deviation of $\overline{Y}$ is $\sigma/\sqrt{N}$: four times the data halves the spread.

In words: averaging N independent measurements keeps the center at the true mean and divides the variance by N, so the typical error of an average shrinks like $1/\sqrt{N}$.

1.4 Conditioning: what $Y \mid X$ and "conditional on X" mean

$Y \mid X$ ("Y given X") is the distribution of Y once we know the value of X. Regression models are written this way: we say how X is drawn, then how Y is drawn *for each value* of X. Lecture 2's example, coding $G = 1$ for male:

$$C \mid G \sim \text{Bernoulli}\big(q_1 G + q_0(1-G)\big),$$

with $q_1 = \mathbb{P}(C{=}1 \mid G{=}1) = 0.00427$ and $q_0 = \mathbb{P}(C{=}1 \mid G{=}0) = 0.00374$. When $G = 1$ the success probability is q_1; when $G = 0$ it is q_0. The difference $q_1 - q_0 = 0.00053$ is the "effect" of being male.

$\mathbb{E}[Y \mid X]$ ("expected Y given X") is the average of Y within the group that has that value of X. In data: `np.mean(y[x == 1])`. It depends on X, so it is itself a random variable.

Tower property: $\mathbb{E}\big[\mathbb{E}[Y \mid X]\big] = \mathbb{E}[Y]$: average the group averages, weighting each group by how likely it is.

Example (Lecture 5): the tower property

$X \sim \text{Bernoulli}(\frac{1}{2})$ and $Y \in \{1,2,3\}$ with $\mathbb{P}(Y \mid X{=}0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ and $\mathbb{P}(Y \mid X{=}1) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Then

$$\mathbb{E}[Y \mid X{=}0] = 1 \cdot \tfrac{1}{2} + 2 \cdot \tfrac{1}{4} + 3 \cdot \tfrac{1}{4} = \tfrac{7}{4}, \qquad \mathbb{E}[Y \mid X{=}1] = \tfrac{1+2+3}{3} = 2,$$
$$\mathbb{E}[Y] = \tfrac{1}{2} \cdot \tfrac{7}{4} + \tfrac{1}{2} \cdot 2 = \tfrac{15}{8}, \qquad \mathbb{E}[XY] = \mathbb{P}(X{=}1) \cdot \mathbb{E}[Y \mid X{=}1] = \tfrac{1}{2} \cdot 2 = 1$$

(when $X = 0$ the product XY is 0, so only the $X = 1$ group contributes).

"Conditional on $X_1, \ldots, X_N$" (used from Section 8 on) means: treat the X values in the dataset

as fixed numbers, and imagine redrawing only the Y's. In the graders example, it means keeping the same assignment of exams to graders and asking how the scores could have come out differently.

1.5 Covariance and correlation

$\operatorname{cov}(X,Y) = \mathbb{E}\big[(X-\mathbb{E}X)(Y-\mathbb{E}Y)\big]$: positive if X and Y tend to be above their means together, negative if one tends to be above when the other is below. Expanding the product and using linearity (this exact computation is on the 10/7 slides):

$$\begin{aligned}\mathbb{E}\big[(X-\mathbb{E}X)(Y-\mathbb{E}Y)\big] &= \mathbb{E}\big[XY - \mathbb{E}[Y]X - \mathbb{E}[X]Y + \mathbb{E}[X]\mathbb{E}[Y]\big] && \text{(expand)}\\ &= \mathbb{E}[XY] - \mathbb{E}[Y]\mathbb{E}[X] - \mathbb{E}[X]\mathbb{E}[Y] + \mathbb{E}[X]\mathbb{E}[Y] && \text{(linearity } (\mathbb{E}X, \mathbb{E}Y \text{ constants}))\\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].\end{aligned}$$

Rules you will need:

- $\operatorname{cov}(X,X) = \operatorname{var}(X)$ and $\operatorname{cov}(X,Y) = \operatorname{cov}(Y,X)$.
- **Bilinearity:** $\operatorname{cov}(X,\, aY+bZ+c) = a\operatorname{cov}(X,Y) + b\operatorname{cov}(X,Z)$. Constants drop out; coefficients come out.
- If X, Y are independent then $\operatorname{cov}(X,Y) = 0$ (because $\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y$). The converse is false.
- In general $\operatorname{var}(X+Y) = \operatorname{var} X + \operatorname{var} Y + 2\operatorname{cov}(X,Y)$.
- **Correlation:** $\rho = \operatorname{corr}(X,Y) = \dfrac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y}$; it has no units and lies in $[-1,1]$.

Example (Lecture 7): reading covariance off a plot

$X \sim \text{Bernoulli}(\frac{1}{2})$ and the plot shows group means $\mathbb{E}[Y \mid X{=}0] \approx 2$, $\mathbb{E}[Y \mid X{=}1] \approx 6$, with spread $\sigma_\epsilon \approx 1$ in each group. The slope is the difference of group means, $\beta_1 \approx 6-2 = 4$. Since $\sigma_X^2 = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$,

$$\operatorname{cov}(X,Y) = \beta_1\sigma_X^2 \approx 4\cdot\tfrac{1}{4} = 1, \qquad \sigma_Y^2 = \beta_1^2\sigma_X^2 + \sigma_\epsilon^2 \approx 16\cdot\tfrac{1}{4} + 1 = 5.$$

(The first formula is proved in Section 1.7; the second in Section 2.)

1.6 Normal, Bernoulli, Binomial

- $X \sim \text{Normal}(\mu,\sigma^2)$: the second number is the *variance*. $aX+b \sim \text{Normal}(a\mu+b, a^2\sigma^2)$, and a sum of independent Normals is Normal (means add, variances add).
- **Standardizing:** $Z = (X-\mu)/\sigma \sim \text{Normal}(0,1)$. We write $\Phi(z) = \mathbb{P}(Z < z)$ for the standard Normal CDF.
- About 68%, 95%, 99.7% of a Normal lies within $1, 2, 3$ standard deviations of the mean. More precisely, 95% lies within $\pm 1.96\,\sigma$; this 1.96 shows up in every confidence interval.
- $Y \sim \text{Bernoulli}(q)$: $\mathbb{E}Y = q$, $\operatorname{var} Y = q(1-q)$. $S = \sum_{i=1}^{N} Y_i \sim \text{Binomial}(N,q)$: $\mathbb{E}S = Nq$, $\operatorname{var} S = Nq(1-q)$.

Example: using the 68% rule

Heights $Y \sim \text{Normal}(64,\ 2.5^2)$ (inches). What fraction are taller than 66.5? Standardize: $z = (66.5 - 64)/2.5 = 1$, so we need $\mathbb{P}(Z > 1)$. 68% lies within one SD, so 32% lies outside, split equally between the two tails by symmetry: $\mathbb{P}(Z > 1) \approx 16\%$ (equivalently $\Phi(1) \approx 0.84$, Lecture 6).

1.7 The single-predictor regression model (from Unit 2)

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X,\ \sigma_\epsilon^2) \qquad \Longleftrightarrow \qquad Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2),$$

with ϵ independent of X. Read it as "to make a Y, take the line's height at X and add Normal noise." The model assumes (1) $Y \mid X$ is Normal, (2) the conditional mean is a line, $\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x$, and (3) the spread $\text{var}(Y \mid X) = \sigma_\epsilon^2$ is the same for every x. Key facts from Unit 2:

$$\mathbb{E}[Y] = \beta_0 + \beta_1 \mathbb{E}[X], \qquad \text{cov}(X, Y) = \beta_1 \sigma_X^2 \implies \beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2}.$$

(Proof of the second: $\text{cov}(X, Y) = \text{cov}(X, \beta_0 + \beta_1 X + \epsilon) = \beta_1 \,\text{cov}(X, X) + \text{cov}(X, \epsilon) = \beta_1 \sigma_X^2 + 0$.)

Example (Lecture 7): from code to a model to a number

```
n = 100000
x = np.random.uniform(0, 2, n)
y = np.zeros(n)
for i in range(n):
    y[i] = np.random.normal(2 + 3*x[i], 1)
print(np.var(y))
```

Model: $X \sim \text{Uniform}(0, 2)$, $Y \mid X \sim \text{Normal}(2 + 3X,\ 1)$ (`np.random.normal` takes the SD, here 1). **What prints?** $Y = 2 + 3X + \epsilon$ with ϵ independent of X, so the variances add: $\text{var}(Y) = 3^2 \,\text{var}(X) + 1$. A Uniform$(a, b)$ has variance $(b - a)^2/12$, so $\text{var}(X) = 4/12 = \frac{1}{3}$ and $\text{var}(Y) = 9 \cdot \frac{1}{3} + 1 = 4$.

2 Correlation and the coefficient of determination (Unit 2.4)

2.1 Correlation is the slope between standardized variables

Let X and Y follow the regression model above and standardize both: $Z_X = (X - \mu_X)/\sigma_X$, $Z_Y = (Y - \mu_Y)/\sigma_Y$. Each standardized variable has mean 0 and standard deviation 1, so it is "in units of its own standard deviation." **Question:** what is the regression coefficient of Z_Y on Z_X? Using $Y = \beta_0 + \beta_1 X + \epsilon$ and $\mu_Y = \beta_0 + \beta_1 \mu_X$:

$$\begin{aligned}
Z_Y = \frac{Y - \mu_Y}{\sigma_Y} &= \frac{(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)}{\sigma_Y} && \text{(substitute } Y \text{ and } \mu_Y\text{)} \\
&= \frac{\beta_1 (X - \mu_X) + \epsilon}{\sigma_Y} && (\beta_0 \text{ cancels; factor out } \beta_1) \\
&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply and divide by } \sigma_X\text{)} \\
&= \frac{\beta_1 \sigma_X}{\sigma_Y} Z_X + \frac{\epsilon}{\sigma_Y}. && \text{(definition of } Z_X\text{)}
\end{aligned}$$

This is again a regression model (no intercept, noise ϵ/σ_Y), with slope $\beta_1\sigma_X/\sigma_Y$. Plugging in $\beta_1 = \text{cov}(X,Y)/\sigma_X^2$:

Correlation

$$\rho = \frac{\beta_1\sigma_X}{\sigma_Y} = \frac{\text{cov}(X,Y)}{\sigma_X\sigma_Y}.$$

ρ is the slope after both variables are measured in standard deviations: if X is one SD above its mean, we expect Y to be ρ SDs above its mean. Since $|\rho| \le 1$, the prediction is always less extreme than X in SD units; this is *regression to the mean.*

In words: correlation is the covariance with the units taken out (divide by both standard deviations); it is the slope you get after measuring X and Y in standard deviations.

2.2 The coefficient of determination ρ^2

Take the variance of both sides of $Y = \beta_0 + \beta_1 X + \epsilon$ (X and ϵ are independent, so their variances add):

$$\begin{aligned}
\sigma_Y^2 &= \beta_1^2\sigma_X^2 + \sigma_\epsilon^2 && \text{(variance of } Y = \beta_0 + \beta_1 X + \epsilon\text{)} \\
1 &= \frac{\beta_1^2\sigma_X^2}{\sigma_Y^2} + \frac{\sigma_\epsilon^2}{\sigma_Y^2} && \text{(divide by } \sigma_Y^2\text{)} \\
1 &= \rho^2 + \frac{\sigma_\epsilon^2}{\sigma_Y^2}. && (\rho = \beta_1\sigma_X/\sigma_Y)
\end{aligned}$$

Coefficient of determination

$$1 - \rho^2 = \frac{\sigma_\epsilon^2}{\sigma_Y^2} = \frac{\text{var}(Y \mid X)}{\text{var}(Y)}, \qquad \text{equivalently} \qquad \rho^2 = 1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2}.$$

ρ^2 is the *fraction of the variance of Y explained by X*. The total variance splits into an explained part $\beta_1^2\sigma_X^2$ (moving along the line) and an unexplained part σ_ϵ^2 (scatter around the line). Since both parts are ≥ 0, $0 \le \rho^2 \le 1$, which is why $-1 \le \rho \le 1$.

In words: $1 - \rho^2$ is the share of Y's variance left over after accounting for X, so ρ^2 is the share X explains.

	Formula	Units	Range
$\text{cov}(X,Y)$	$\mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y$	units $Y\times$ units X	$(-\infty,\infty)$
β_1	$\text{cov}(X,Y)/\sigma_X^2$	units Y / units X	$(-\infty,\infty)$
ρ	$\text{cov}(X,Y)/(\sigma_X\sigma_Y)$	none	$[-1,1]$
ρ^2	$1 - \sigma_\epsilon^2/\sigma_Y^2$	none	$[0,1]$

Summary: measures of association (10/2 slides). Only ρ and ρ^2 can be compared across datasets with different units.

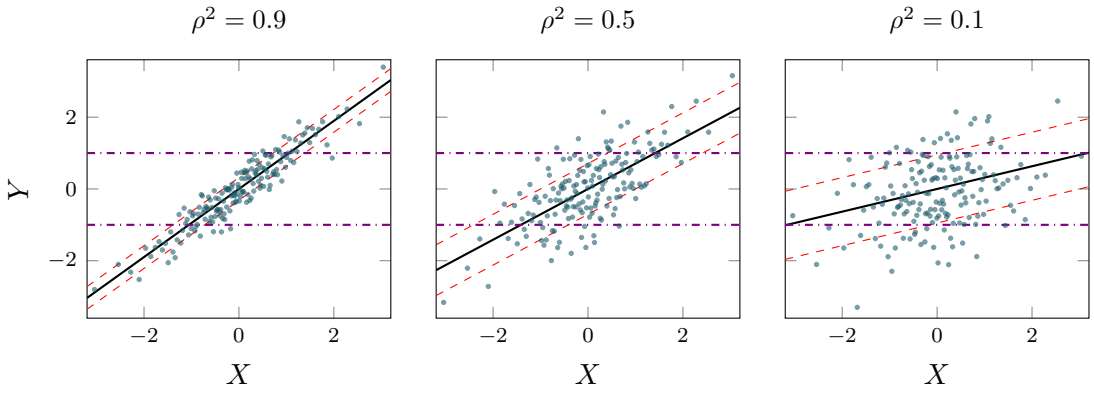


Figure 1: Simulated data with $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$, so $\beta_1 = \rho$. Black: $\mathbb{E}[Y \mid X] = \rho X$. Red dashed: $\pm\sigma_\epsilon = \pm\sqrt{1-\rho^2}$ around the line. Purple: $\pm\sigma_Y = \pm 1$. When ρ^2 is large, the scatter around the line (red) is much narrower than the total scatter (purple); when ρ^2 is small they are almost the same width.

3 The Law of Large Numbers and the Central Limit Theorem (Unit 3.1)

These two theorems explain *why* sample averages work and why Normal distributions show up everywhere in this unit.

Law of Large Numbers (LLN)

If $X_1, \ldots, X_N$ are iid with $\mathbb{E}[X_i] < \infty$ and $S_N = \sum_{i=1}^N X_i$, then $S_N/N \to \mathbb{E}[X_i]$ as $N \to \infty$.

In words: with enough independent data, the sample average lands close to the true mean.

In words: for large N the sample average will not differ from the true mean by much. The toolkit tells us *how* much: $\text{var}(S_N/N) = \sigma^2/N$, so the typical error shrinks like $1/\sqrt{N}$. That calculation never used the shape of the distribution, only its variance, which hints that something universal is going on.

Central Limit Theorem (CLT)

If $X_1, \ldots, X_N$ are iid with $\mathbb{E}[X_i] = \mu$, $\text{var}(X_i) = \sigma^2$, and $S_N = \sum_i X_i$, then

$$\mathbb{P}\left(\frac{S_N - N\mu}{\sqrt{N\sigma^2}} < z\right) \to \mathbb{P}(Z < z) = \Phi(z), \qquad Z \sim \text{Normal}(0,1).$$

Informally: $S_N \approx \text{Normal}(N\mu,\ N\sigma^2)$, and dividing by N, $\overline{Y} \approx \text{Normal}(\mu,\ \sigma^2/N)$, *whatever* the distribution of the X_i (as long as the variance is finite).

In words: add up (or average) many independent pieces and the result follows a bell curve, whatever shape each piece has.

The fraction $(S_N - N\mu)/\sqrt{N\sigma^2}$ is just S_N standardized: subtract its mean $N\mu$, divide by its standard deviation $\sqrt{N\sigma^2}$.

The iid assumption matters. The CLT above needs independent, identically distributed pieces. Strong correlation between the X_i (as in the time series of Section 9) can change the sample distribution a lot, so check that iid is reasonable first.

Example: Binomial approximation (drill from the notes)

$Y \sim \text{Binomial}(200, 0.3)$, i.e. Y is a sum of $N = 200$ iid Bernoulli(0.3) variables. Estimate $\mathbb{P}(Y < 70)$.
Each Bernoulli has $\mu = 0.3$ and $\sigma^2 = 0.3 \cdot 0.7 = 0.21$, so $\mathbb{E}Y = 200(0.3) = 60$ and $\operatorname{var} Y = 200(0.21) = 42$, $\sigma_Y = \sqrt{42} \approx 6.48$. By the CLT, $Y \approx \text{Normal}(60, 42)$, and standardizing,

$$\mathbb{P}(Y < 70) \approx \mathbb{P}\left(Z < \frac{70 - 60}{6.48}\right) = \Phi(1.54) \approx 0.94.$$

The exact Binomial answer is 0.927. The approximation gets better as N grows, because the CLT is a statement about the limit $N \to \infty$.

```
from scipy.stats import norm, binom
print(norm.cdf((70 - 60) / np.sqrt(42)))   # CLT approximation, about 0.939
print(binom.cdf(69, 200, 0.3))             # exact P(Y <= 69) = P(Y < 70), about 0.927
```

4 Estimators and the sample distribution (Unit 3.2)

4.1 Estimators

Estimator

An **estimator** $\hat{\theta}$ of a parameter θ is any function of the data meant to approximate θ.

- $\hat{\theta}$ is **random**: it changes depending on which sample we collect.
- θ is a **fixed, non-random number** (in frequentist statistics; this changes in Unit 6).

Examples we have already met: $\hat{q} = \overline{Y}$ for a Bernoulli probability, $\hat{\mu} = \overline{Y}$ for a Normal mean, $\hat{\beta}_1 = \bar{Y}_{X=1} - \bar{Y}_{X=0}$ for a binary-predictor regression.

4.2 Replicates and the sample distribution

A **replicate** is a whole new copy of the dataset (repeat the survey, rerun the experiment). Be careful with the words: a replicate *consists of* many samples, and each replicate produces one value of $\hat{\theta}$.

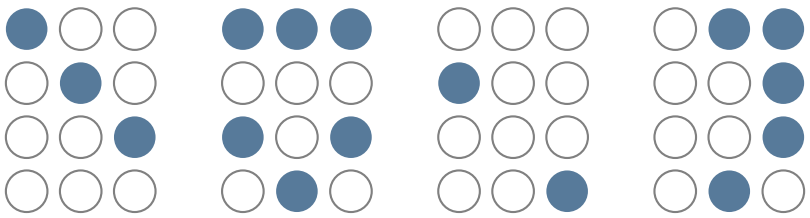


Figure 2: Four replicates of a survey of $N = 12$ people (filled = YES). The true q never changes, but $\hat{q}$ changes from replicate to replicate. The distribution of $\hat{q}$ over (infinitely many) replicates is its **sample distribution**; its spread is $\text{se}(\hat{q})$.

Sample distribution and standard error

The **sample distribution** of $\hat{\theta}$ is its distribution over replicates of the data. The **standard error** $\text{se}(\hat{\theta})$ is the standard deviation of the sample distribution. It answers: "if I redid the whole study, by how much would my estimate typically change?"

Example: the sample distribution of a Normal mean

Let $Y_1, \ldots, Y_N \sim \text{Normal}(\mu, \sigma^2)$ iid and estimate μ with $\hat{\mu} = \overline{Y} = \frac{1}{N}\sum_i Y_i$.
$\sum_i Y_i$ is a sum of independent Normals, so it is exactly Normal with mean $N\mu$ and variance $N\sigma^2$. Dividing by N (which divides the mean by N and the variance by N^2):

$$\hat{\mu} \sim \text{Normal}\left(\mu,\ \frac{\sigma^2}{N}\right), \qquad \text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}.$$

For non-Normal data with finite variance, the CLT gives the same statement as a large-N approximation.

True vs. estimated standard error. The formula $\sigma/\sqrt{N}$ contains σ, which in real life we do not know. So we plug in an estimate, $\text{se}(\hat{\mu}) = \hat{\sigma}/\sqrt{N}$. If we are working with a model whose parameters are given, "standard error" means the first formula; with real data it means the second. Some people call the distribution with $\hat{\sigma}$ plugged in the sample distribution too.

Example: estimating q (10/2 slides)

Let $Y_1, \ldots, Y_N \sim \text{Bernoulli}(q)$ and $\hat{q} = \overline{Y}$. Now $\hat{q}$ is *not* Normal (it can only take the values $0, \frac{1}{N}, \ldots, 1$), but by the CLT the Normal result carries over for large N, with $\sigma^2 = q(1-q)$:

$$\hat{q} \approx \text{Normal}\left(q,\ \frac{q(1-q)}{N}\right), \qquad \text{se}(\hat{q}) = \sqrt{\frac{q(1-q)}{N}}.$$

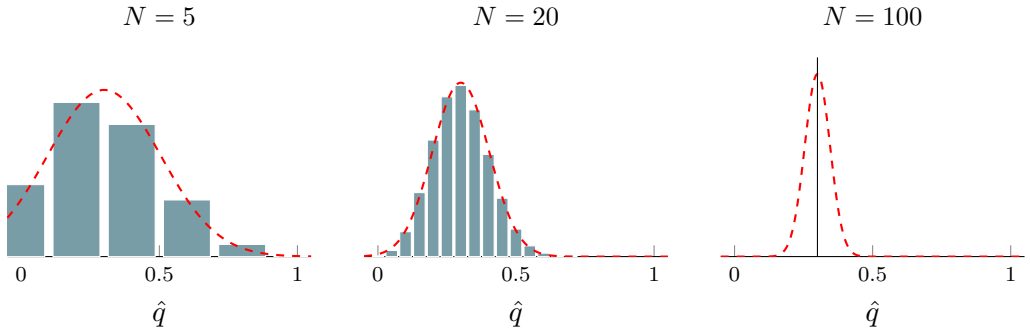


Figure 3: Sample distribution of $\hat{q} = \overline{Y}$ for $q = 0.3$. Bars: the exact distribution (scaled to a density). Red dashed: the CLT approximation $\text{Normal}(q, q(1-q)/N)$. As N grows, the bars look more and more Normal and the distribution concentrates around $q = 0.3$.

5 Bias, consistency, and the bias–variance trade-off (Unit 3.2)

5.1 Definitions

MSE, bias, unbiased, consistent

- **Mean squared error:** $\mathrm{MSE}_{\hat\theta} = \mathbb{E}\big[(\hat\theta - \theta)^2\big]$, the average squared miss over replicates.
- **Bias:** $\mathrm{Bias} = \mathbb{E}[\hat\theta] - \theta$. $\hat\theta$ is **unbiased** if the bias is 0 for every N (it is right on average).
- **Consistent:** $\mathbb{E}[\hat\theta] \to \theta$ *and* $\mathrm{se}(\hat\theta) \to 0$ as $N \to \infty$ (more data pins down θ).

Bias–variance decomposition

$$\mathrm{MSE}_{\hat\theta} = \mathrm{se}(\hat\theta)^2 + \big(\mathbb{E}[\hat\theta] - \theta\big)^2 = \text{variance} + \text{bias}^2.$$

In words: the average squared error of an estimate is how much it jitters from dataset to dataset (variance) plus how far off it is on average (bias), squared.

Proof. Apply the variance formula $\mathrm{var}(W) = \mathbb{E}[W^2] - (\mathbb{E}W)^2$ to $W = \hat\theta - \theta$:

$$\begin{aligned} \mathrm{var}(\hat\theta - \theta) &= \mathbb{E}\big[(\hat\theta - \theta)^2\big] - \big(\mathbb{E}[\hat\theta - \theta]\big)^2 && \text{(variance formula)} \\ \mathrm{var}(\hat\theta) &= \mathrm{MSE}_{\hat\theta} - \big(\mathbb{E}[\hat\theta] - \theta\big)^2. && (\theta \text{ is a constant: shifting does not change variance}) \end{aligned}$$

Rearranging gives the result. □

So an estimator can be wrong in two ways: it can be *noisy* (large variance) or *systematically off* (bias). The picture to keep in mind is a dartboard: bias is how far the center of your throws is from the bullseye; variance is how spread out they are.

Unbiased and consistent are different

$\hat\mu_1 = \overline{Y} + 1/N$ is biased (bias $= 1/N$) but consistent. $\hat\mu_2 = (Y_1 + Y_2)/2$ is unbiased ($\mathbb{E}\hat\mu_2 = \mu$) but *not* consistent: $\hat\mu_2 \sim \mathrm{Normal}(\mu, \sigma^2/2)$, and its standard error $\sigma/\sqrt{2}$ never shrinks, because it only ever uses two data points.

Question (10/5): is the sample mean unbiased? consistent?

$\mathbb{E}[\overline{Y}] = q$ (Bernoulli) or μ (Normal), by the toolkit box, so it is unbiased. $\mathrm{se}(\overline{Y}) = \sqrt{q(1-q)/N}$ or $\sigma/\sqrt{N}$, both $\to 0$ as $N \to \infty$, so it is consistent.

5.2 Laplace's rule of succession

The natural estimator of q is $\hat q = \overline{Y}$. But if every $Y_i = 0$ then $\hat q = 0$: flip a coin twice, get two tails, and conclude the coin can *never* land heads. To fix this, pretend we have already seen two extra flips, one heads and one tails:

$$\hat q_L = \frac{S+1}{N+2}, \qquad S = \sum_i Y_i \sim \mathrm{Binomial}(N, q).$$

Question (10/5): find the bias, variance and MSE of $\hat{q}_L$

Mean. Using $\mathbb{E}[S] = Nq$ and linearity,

$$\mathbb{E}[\hat{q}_L] = \frac{\mathbb{E}[S]+1}{N+2} = \frac{Nq+1}{N+2}.$$

Bias. Put both terms over the common denominator $N+2$:

$$\text{Bias} = \frac{Nq+1}{N+2} - q = \frac{Nq+1-q(N+2)}{N+2} = \frac{Nq+1-Nq-2q}{N+2} = \frac{1-2q}{N+2}.$$

Variance. $\hat{q}_L = \frac{1}{N+2}S + \frac{1}{N+2}$ is $aS+b$ with $a = \frac{1}{N+2}$, so

$$\text{var}(\hat{q}_L) = \frac{\text{var}(S)}{(N+2)^2} = \frac{Nq(1-q)}{(N+2)^2}.$$

MSE = variance + bias^2:

$$\text{MSE}_{\hat{q}_L} = \frac{Nq(1-q)+(1-2q)^2}{(N+2)^2}.$$

Verdict. Biased (unless $q = \frac{1}{2}$, where $1-2q=0$), but consistent: the bias has N in the denominator, and the variance behaves like $N/N^2 = 1/N$; both go to 0.
Is the bias worth it? Compare with $\overline{Y}$, whose MSE is just its variance $q(1-q)/N$. At $N=2$, $q=\frac{1}{2}$: $\text{MSE}_{\overline{Y}} = 0.25/2 = 0.125$ but $\text{MSE}_{\hat{q}_L} = 2(0.25)/16 = 0.031$. Laplace's estimator is four times more accurate there, even though it is biased. At $q=0$ it loses ($\overline{Y}$ is then always exactly right). Unbiased is not automatically best.

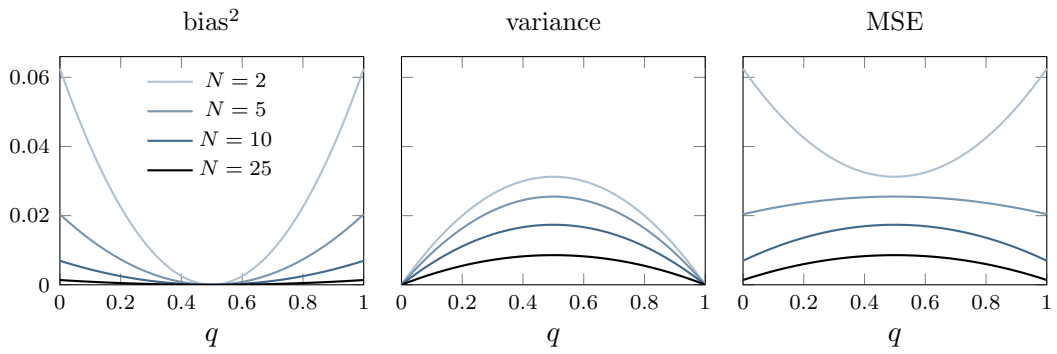


Figure 4: Bias^2, variance and MSE of Laplace's estimator, from the formulas above. Bias is largest near $q=0$ and $q=1$ (the estimator is pulled toward $\frac{1}{2}$); variance is largest at $q=\frac{1}{2}$. All three shrink as N grows.

The slides check these formulas by simulation. Each pass through the inner loop is one replicate of the data, so `q_hat` holds R draws from the sample distribution of $\hat{q}_L$. Then the sample bias^2, variance and MSE of those draws estimate the true ones:

```
N = 10
R = 1000                        # number of replicates
qs = np.linspace(0, 1, 11)
for i in range(len(qs)):
    q = qs[i]
    q_hat = np.zeros(R)
```

```
    for r in range(R):
        S = np.random.binomial(N, q)
        # This also works:
        # ys = np.random.choice([0, 1], N, p=[1-q, q])
        # S = np.sum(ys)
        q_hat[r] = (S + 1) / (N + 2)
    bias2[i] = (np.mean(q_hat) - q)**2
    var[i] = np.var(q_hat)
    mse[i] = np.mean((q_hat - q)**2)
```

5.3 Estimating the variance: Bessel's correction

Question (10/5): μ known

If $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and we *know* μ, is $\hat{\sigma}^2_\mu = \frac{1}{N}\sum_i (Y_i - \mu)^2$ unbiased?
Yes: $\mathbb{E}[(Y_i-\mu)^2] = \text{var}(Y_i) = \sigma^2$ by the definition of variance, so $\mathbb{E}[\hat{\sigma}^2_\mu] = \frac{1}{N}\sum_i \sigma^2 = \frac{1}{N}\cdot N\sigma^2 = \sigma^2$.

Usually μ is unknown, so we use $\overline{Y}$ in its place: $\hat{\sigma}^2_0 = \frac{1}{N}\sum_i (Y_i - \overline{Y})^2$. This one is **biased**. The trick to see why: add and subtract μ inside the square, then expand $(a-b)^2 = a^2 - 2ab + b^2$ with $a = Y_i - \mu$ and $b = \overline{Y} - \mu$:

$$\begin{aligned}
\sum_i (Y_i - \overline{Y})^2 &= \sum_i \big[(Y_i - \mu) - (\overline{Y} - \mu)\big]^2 && \text{(add and subtract } \mu\text{)} \\
&= \sum_i (Y_i - \mu)^2 - 2(\overline{Y} - \mu)\sum_i (Y_i - \mu) + N(\overline{Y} - \mu)^2 && \text{(expand; } \overline{Y} - \mu \text{ same for all } i\text{)} \\
&= \sum_i (Y_i - \mu)^2 - 2N(\overline{Y} - \mu)^2 + N(\overline{Y} - \mu)^2 && \text{(} \textstyle\sum_i (Y_i - \mu) = N(\overline{Y} - \mu)\text{)} \\
&= \sum_i (Y_i - \mu)^2 - N(\overline{Y} - \mu)^2.
\end{aligned}$$

Now take expectations. $\mathbb{E}[(Y_i - \mu)^2] = \sigma^2$, and $\mathbb{E}[(\overline{Y} - \mu)^2] = \text{var}(\overline{Y}) = \sigma^2/N$ (because $\mathbb{E}\overline{Y} = \mu$). So

$$\mathbb{E}\Big[\sum_i (Y_i - \overline{Y})^2\Big] = N\sigma^2 - N\cdot\frac{\sigma^2}{N} = (N-1)\sigma^2, \qquad \text{so} \qquad \mathbb{E}[\hat{\sigma}^2_0] = \frac{N-1}{N}\sigma^2.$$

Why too small? $\overline{Y}$ is computed from the same data, so it sits closer to the data points than the true μ does. Measuring spread around $\overline{Y}$ therefore underestimates spread around μ; the bias is $-\sigma^2/N$.

Bessel's correction

Dividing by $N-1$ instead of N removes the bias:

$$\hat{\sigma}^2 = \frac{1}{N-1}\sum_i (Y_i - \overline{Y})^2, \qquad \mathbb{E}[\hat{\sigma}^2] = \sigma^2.$$

$\hat{\sigma}^2_0$ is biased but still consistent (its bias $-\sigma^2/N \to 0$). For large N the two barely differ.

In words: when the mean is estimated from the same data, divide the sum of squared deviations by $N-1$ instead of N to get the variance right on average.

Check with $N = 2$

With two points, $\overline{Y} = (Y_1 + Y_2)/2$, so $Y_1 - \overline{Y} = (Y_1 - Y_2)/2$ and $Y_2 - \overline{Y} = -(Y_1 - Y_2)/2$. Then $\hat{\sigma}_0^2 = \frac{1}{2}\left[\frac{(Y_1-Y_2)^2}{4} + \frac{(Y_1-Y_2)^2}{4}\right] = \frac{(Y_1-Y_2)^2}{4}$. Since $\mathbb{E}[Y_1 - Y_2] = 0$, $\mathbb{E}[(Y_1 - Y_2)^2] = \text{var}(Y_1 - Y_2) = \sigma^2 + \sigma^2 = 2\sigma^2$, so $\mathbb{E}[\hat{\sigma}_0^2] = \sigma^2/2$, which matches $\frac{N-1}{N}\sigma^2$ at $N = 2$.

6 Confidence intervals (Unit 3.3)

6.1 True vs. estimated sample distribution

For $Y_i \sim \text{Normal}(\mu, \sigma^2)$, the *true* sample distribution of $\hat{\mu} = \overline{Y}$ is $\text{Normal}(\mu, \sigma^2/N)$. But we know neither μ nor σ! So we plug in our estimates to get the **estimated sample distribution**

$$\hat{\mu} \approx \text{Normal}\left(\hat{\mu},\ \frac{\hat{\sigma}^2}{N}\right) \qquad \text{(red: numbers computed from our data).}$$

This estimated distribution is itself random (it is centered at $\hat{\mu}$, not μ, and its width uses $\hat{\sigma}$). Still, its width tells us how far μ might plausibly be from $\hat{\mu}$. Confidence intervals make this precise.

6.2 The 95% confidence interval

95% confidence interval

$$\text{CI} = \left[\hat{\theta} - 1.96\ \text{se}(\hat{\theta}),\quad \hat{\theta} + 1.96\ \text{se}(\hat{\theta})\right].$$

In words: go 1.96 standard errors either side of the estimate; intervals built this way contain the true value in 95% of repeated studies.

Why 1.96? If the sample distribution is approximately Normal, 95% of it lies within 1.96 standard errors of its center. So in 95% of replicates $\hat{\theta}$ lands within 1.96 se of θ, which is the same as saying θ lands within 1.96 se of $\hat{\theta}$, i.e. inside the CI.

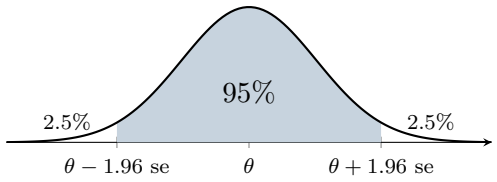


Figure 5: An (approximately) Normal sample distribution. 95% of replicates give $\hat{\theta}$ within 1.96 standard errors of θ.

The t multiplier. With $\hat{\sigma}$ plugged in, $(\hat{\mu} - \mu)/\,\text{se}(\hat{\mu})$ is not exactly $\text{Normal}(0, 1)$; it has a t **distribution**, which looks Normal but has heavier tails (because $\hat{\sigma}$ is random too). The exact 95% multiplier is then a bit bigger than 1.96: about 2.78 for $N = 5$, 2.26 for $N = 10$, 2.05 for $N = 30$, and ≈ 1.96 for large N. For $N \gtrsim 30$ the difference is negligible. Software such as statsmodels reports t-based numbers, so they may differ slightly from yours.

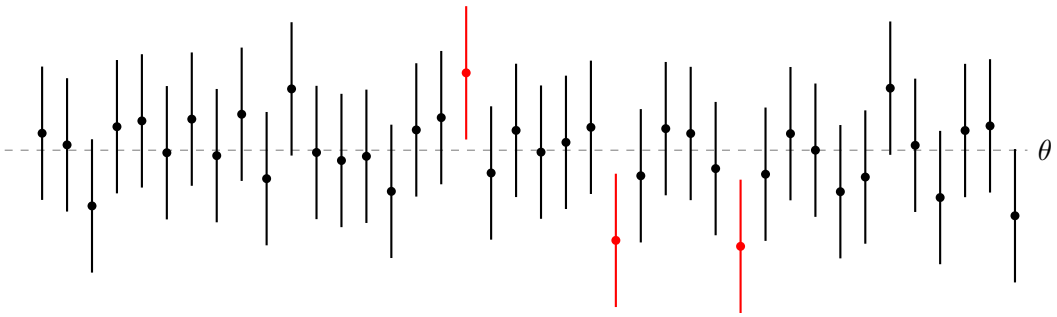


Figure 6: Each bar is the 95% CI from one simulated replicate, with a dot at $\hat{\theta}$. The CIs move around; θ (dashed) does not. About 95% of the bars cover θ; the red ones miss.

6.3 Interpreting a CI correctly (optional subtlety)

- **Correct:** if we generate many replicates of the data, the true θ falls in the resulting CI for 95% of them; that is, $\mathbb{P}(\theta \in \text{CI}) = 0.95$ where the CI is treated as a *random* interval.
- **Not correct:** "there is a 95% chance θ lies in *this particular* interval." Once we compute, say, $\text{CI} = [0.2, 1.4]$, nothing is random any more: θ is either in it or not, so $\mathbb{P}(\theta \in \text{CI})$ is 0 or 1.

For practical purposes you can think of the CI as "the region where the parameter is likely to be."

Example (notes): how much data for a target CI width?

Model $Y_i \sim \text{Normal}(\mu, \sigma^2)$, estimate μ by $\overline{Y}$, and from earlier experiments $\sigma \approx 1$. How large must N be for the 95% CI $\hat{\mu} \pm 1.96\ \text{se}(\hat{\mu})$ to be no wider than $\hat{\mu} \pm 0.25$?
The half-width is $1.96\ \text{se} = 1.96/\sqrt{N}$. We need

$$\frac{1.96}{\sqrt{N}} \le 0.25 \implies \sqrt{N} \ge \frac{1.96}{0.25} = 7.84 \implies N \ge 7.84^2 \approx 61.5,$$

so $N = 62$. (Since $\sigma \approx 1$ is itself an estimate, this is approximate.) Notice the square: halving the width needs four times the data.

7 Hypothesis testing (Unit 3.6)

Often we estimate a parameter to make a *decision*: is a drug worth pursuing? Do two graders grade differently? The procedure:

1. State a **null hypothesis** H_0 we might want to rule out, e.g. $H_0 : \beta_1 = 0$ ("no effect").
2. Compute a **test statistic** $\hat{T}$ from the data. For a regression slope we use how many standard errors the estimate is from 0:
$$\hat{T} = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)}.$$
3. Ask: if H_0 were true, how likely is a test statistic at least as extreme as ours? Let $\hat{T}^*$ be the test statistic from a replicate generated under H_0. Then $\hat{T}^* \approx \text{Normal}(0, 1)$, and the **$p$-value** is
$$p_v = \mathbb{P}\big(|\hat{T}^*| > |\hat{T}|\big).$$

A small p_v means our data would be very surprising if H_0 were true, so we *reject* H_0. A large p_v means data like ours are typical under H_0, so we *cannot reject* it (which is not the same as proving H_0 true). Note the connection to CIs: $|\hat{T}| > 1.96$ exactly when the 95% CI does not contain 0, and then $p_v < 0.05$.

Example (notes): a clinical trial is a binary-predictor regression

Effect Y in a control group ($X = 0$) and a treatment group ($X = 1$), $N/2$ people each, with σ_ϵ known: $Y \mid X \sim \text{Normal}\big(\mu_C(1-X) + \mu_T X,\ \sigma_\epsilon^2\big)$. Since $\mu_C(1-X) + \mu_T X = \mu_C + (\mu_T - \mu_C)X$, this is a regression with $\beta_0 = \mu_C$ and $\beta_1 = \mu_T - \mu_C$.
Each group mean is the average of $N/2$ Normals, so $\hat{\mu}_C \sim \text{Normal}(\mu_C, \frac{\sigma_\epsilon^2}{N/2})$ and likewise for $\hat{\mu}_T$. The two groups are independent, so variances add for the difference:

$$\hat{\beta}_1 = \hat{\mu}_T - \hat{\mu}_C \sim \text{Normal}\left(\beta_1,\ \frac{2\sigma_\epsilon^2}{N} + \frac{2\sigma_\epsilon^2}{N}\right) = \text{Normal}\left(\beta_1,\ \frac{4\sigma_\epsilon^2}{N}\right).$$

Under $H_0 : \beta_1 = 0$ the replicate $\hat{\beta}_1^* \sim \text{Normal}(0, 4\sigma_\epsilon^2/N)$, so $\hat{T}^* = \hat{\beta}_1^* / \text{se}(\hat{\beta}_1) \sim \text{Normal}(0, 1)$.

Worked example (10/7): do the two midterm graders grade differently?

Midterm 1 was graded by E ($X = 1$, $N_E = 23$ exams) and G ($X = 0$, $N_G = 45$). Model: $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$, so β_1 is the difference in average score between graders.
Sample distribution of $\hat{\beta}_1$ ("eyeball" numbers from the plot). $\overline{Y|X{=}0} \approx 17$, $\overline{Y|X{=}1} \approx 17.5$, $\hat{\sigma}_\epsilon \approx 5$. Each group mean is approximately Normal with variance $\hat{\sigma}_\epsilon^2/(\text{group size})$:

$$\overline{Y|X{=}1} \approx \text{Normal}(17.5,\ 25/23), \qquad \overline{Y|X{=}0} \approx \text{Normal}(17,\ 25/45).$$

The groups are independent, so for the difference the means subtract and the variances add. Using $25/23 + 25/45 \approx 1 + \frac{1}{2}$:

$$\hat{\beta}_1 = \overline{Y|X{=}1} - \overline{Y|X{=}0} \approx \text{Normal}\big(\tfrac{1}{2},\ \tfrac{3}{2}\big).$$

95% CI. $\text{se}(\hat{\beta}_1) \approx \sqrt{3/2} \approx 1.2$. Using $1.96 \approx 2$: CI $\approx [\frac{1}{2} - 2(1.2),\ \frac{1}{2} + 2(1.2)] \approx [-2, 3]$, which contains 0.
Test statistic. $\hat{T} = \hat{\beta}_1 / \text{se}(\hat{\beta}_1) \approx 0.5/1.2 \approx 0.4$: the difference is only 0.4 standard errors from 0.
***p*-value by hand.** Near its center the standard Normal density is about $1/\sqrt{2\pi} \approx 0.4$, so the area between -0.4 and 0.4 is roughly a rectangle of width 0.8 and height 0.4: $\mathbb{P}(|\hat{T}^*| < 0.4) \approx 0.32$. Hence $p_v = \mathbb{P}(|\hat{T}^*| > 0.4) \approx 0.7$ (the exact value $2(1 - \Phi(0.4))$ is 0.69). A difference like ours is completely typical if the graders grade the same, so we *cannot reject* H_0.
In Python (with `x`, `y` numpy arrays) the exact numbers are:

```
y0 = y[x == 0]
y1 = y[x == 1]
b1_hat = np.mean(y1) - np.mean(y0)
var_b1 = np.std(y1)**2/len(y1) + np.std(y0)**2/len(y0)
```

giving $\hat{\beta}_1 \approx \text{Normal}(0.68,\ 1.84)$, close to the eyeball answer.

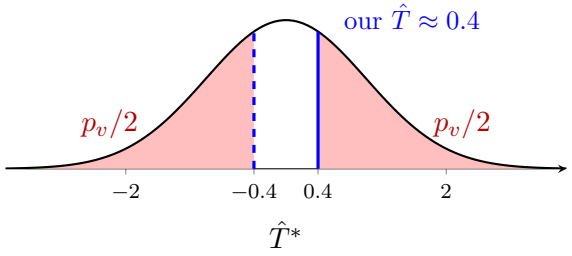


Figure 7: The null distribution of $\hat{T}^*$. The p-value is the shaded area beyond $\pm|\hat{T}|$: here about 0.7, so most of the curve.

7.1 p-values are not R^2

The test statistic and R^2 (one predictor)

$$\hat{T}^2 = (N-2)\,\frac{R^2}{1-R^2}.$$

In words: with one predictor, $\hat{T}$ depends only on R^2 and N: a high R^2 always gives a big $|\hat{T}|$ (tiny p-value), but a tiny p-value can come with a low R^2 when N is large.

Derivation (uses Section 8; skip on a first read). Write $S_{xx} = \sum_i (X_i - \overline{X})^2$. The residual is $r_i = (Y_i - \overline{Y}) - \hat{\beta}_1(X_i - \overline{X})$ (because $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1\overline{X}$). Squaring and summing,

$$\begin{aligned}\mathrm{SSR} &= \sum_i (Y_i - \overline{Y})^2 - 2\hat{\beta}_1 \sum_i (X_i - \overline{X})(Y_i - \overline{Y}) + \hat{\beta}_1^2 S_{xx} \\ &= \mathrm{SST} - 2\hat{\beta}_1(\hat{\beta}_1 S_{xx}) + \hat{\beta}_1^2 S_{xx} = \mathrm{SST} - \hat{\beta}_1^2 S_{xx} \qquad \left(\sum_i (X_i - \overline{X})(Y_i - \overline{Y}) = \hat{\beta}_1 S_{xx}\right)\end{aligned}$$

so $\hat{\beta}_1^2 S_{xx} = \mathrm{SST} - \mathrm{SSR}$. With $\mathrm{se}(\hat{\beta}_1)^2 = \hat{\sigma}_\epsilon^2 / S_{xx} = \mathrm{SSR}/((N-2)S_{xx})$,

$$\hat{T}^2 = \frac{\hat{\beta}_1^2}{\mathrm{se}(\hat{\beta}_1)^2} = \frac{(N-2)\,\hat{\beta}_1^2 S_{xx}}{\mathrm{SSR}} = \frac{(N-2)(\mathrm{SST} - \mathrm{SSR})}{\mathrm{SSR}} = (N-2)\frac{1 - \mathrm{SSR}/\mathrm{SST}}{\mathrm{SSR}/\mathrm{SST}} = (N-2)\frac{R^2}{1-R^2}.$$

Question (10/9): which scenarios are possible? (p_v is for $H_0 : \beta_1 = 0$, $N > 3$)

(a) $p_v = 10^{-5}$ and $R^2 = 10^{-2}$: possible. R^2 small means X explains little of Y; but with a huge N even a tiny slope is measured precisely. From the formula, $\hat{T}^2 \approx 0.0101(N-2)$; $p_v = 10^{-5}$ needs $|\hat{T}| \approx 4.4$, i.e. $N \approx 1950$.
(b) $p_v = 0.5$ and $R^2 = 0.99$: impossible. $R^2 = 0.99$ forces $\hat{T}^2 = 99(N-2) \geq 198$, so $|\hat{T}| \geq 14$ and p_v is tiny.
(c) $\hat{\beta}_1 = 1000$, $\hat{\sigma}_\epsilon^2 = 10^{-5}$, $R^2 = 10^{-4}$: possible. A big slope does not mean a big R^2: $R^2 \approx \beta_1^2\sigma_X^2/(\beta_1^2\sigma_X^2 + \sigma_\epsilon^2)$, which is tiny if X barely varies ($\sigma_X \approx 3 \times 10^{-8}$).

Lessons from the "guess R^2 and p_v" plots (10/9).

- A U-shaped pattern can have $R^2 \approx 0$ and a large p_v: there is a strong relationship, just not a *linear* one.
- A noisy cloud with a slight trend and many points can have small R^2 (0.10) but a tiny p_v (3×10^{-6}).

- Real data with a clear trend (dog weight vs. life expectancy, $R^2 = 0.44$; bitcoin log price vs. year, 0.86; CO_2, 0.98) have astronomically small p_v. A tiny p_v says the slope is not 0; it does not say the line is a good model (the CO_2 data curve).

8 Least squares for single-predictor regression (Unit 3.4)

8.1 The estimators

Data $(X_1, Y_1), \dots, (X_N, Y_N)$ from $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$. How do we estimate β_1? We know $\beta_1 = \text{cov}(X, Y)/\sigma_X^2$, and both are expectations: $\text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, $\sigma_X^2 = \mathbb{E}[(X - \mathbb{E}X)^2]$. Replace each expectation by a sample average, $\mathbb{E}[\,\cdot\,] \to \frac{1}{N}\sum_i$, and $\mathbb{E}X, \mathbb{E}Y$ by $\overline{X}, \overline{Y}$:

$$\text{cov}(X, Y) \approx \frac{1}{N}\sum_i (X_i - \overline{X})(Y_i - \overline{Y}), \qquad \sigma_X^2 \approx \frac{1}{N}\sum_i (X_i - \overline{X})^2.$$

The $\frac{1}{N}$'s cancel in the ratio. For the intercept, $\mathbb{E}Y = \beta_0 + \beta_1 \mathbb{E}X$ gives $\beta_0 = \mathbb{E}Y - \beta_1 \mathbb{E}X$.

Least squares estimators

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}, \qquad \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}.$$

In words: the slope is the sample covariance of X and Y divided by the sample variance of X; the intercept makes the line pass through the point of averages $(\overline{X}, \overline{Y})$.

The fitted line always passes through the point $(\overline{X}, \overline{Y})$: plug $X = \overline{X}$ into $\hat{\beta}_0 + \hat{\beta}_1 X$ to get $\overline{Y}$.

8.2 Residuals, SSR, SST, $\hat{\sigma}_\epsilon^2$, R^2

The **residuals** are the vertical distances from each point to the fitted line, $r_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$.

$$\text{SSR} = \sum_i r_i^2 \ \text{(sum of squared residuals)}, \qquad \text{SST} = \sum_i (Y_i - \overline{Y})^2 \ \text{(total sum of squares)}.$$

(Careful with other books: some use "SSR" for the *regression* sum of squares SST − SSE, a different thing.)

Estimators of the noise variance and of ρ^2

$$\hat{\sigma}_\epsilon^2 = \frac{\text{SSR}}{N - 2} \approx \frac{\text{SSR}}{N}, \qquad \hat{\rho}^2 = R^2 = 1 - \frac{\text{SSR}}{\text{SST}}.$$

In words: $\hat{\sigma}_\epsilon^2$ is the average squared vertical distance from the points to the line (dividing by $N - 2$); R^2 is the fraction of the total spread in Y that the line accounts for.

Why these make sense. SSR/N is the average squared scatter around the line, a sample version of $\sigma_\epsilon^2 = \text{var}(Y \mid X)$. SST/N is the sample version of σ_Y^2. So $1 - \text{SSR}/\text{SST}$ copies $\rho^2 = 1 - \sigma_\epsilon^2/\sigma_Y^2$. **Why $N - 2$?** The same reason as Bessel's correction: two parameters (β_0, β_1) were fitted from the same data, which makes the line sit a bit too close to the points; dividing by $N - 2$ makes $\hat{\sigma}_\epsilon^2$ unbiased. It makes little difference once $N \gtrsim 20$.

8.3 Why "least squares"

Least squares property

$$(\hat\beta_0, \hat\beta_1) = \operatorname*{argmin}_{b_0, b_1} \sum_i \big(Y_i - b_0 - b_1 X_i\big)^2 :$$

of all lines, the least squares line has the smallest SSR. This holds for any set of points, wherever they came from.

In words: of every possible straight line, the least squares line makes the total of the squared vertical misses as small as possible.

Optional calculus check. Call $f(b_0, b_1) = \sum_i (Y_i - b_0 - b_1 X_i)^2$. At the minimum, changing b_0 alone (holding b_1 fixed) cannot lower f, so the ordinary derivative in b_0 is 0; same for b_1. By the chain rule:

$$\text{in } b_0: \quad \sum_i -2(Y_i - b_0 - b_1 X_i) = 0 \implies N\overline{Y} - Nb_0 - b_1 N\overline{X} = 0 \implies b_0 = \overline{Y} - b_1\overline{X},$$

$$\text{in } b_1: \quad \sum_i -2X_i(Y_i - b_0 - b_1 X_i) = 0.$$

Substitute $b_0 = \overline{Y} - b_1\overline{X}$ into the second equation: $\sum_i X_i\big[(Y_i - \overline{Y}) - b_1(X_i - \overline{X})\big] = 0$, so $b_1 = \dfrac{\sum_i X_i(Y_i - \overline{Y})}{\sum_i X_i(X_i - \overline{X})}$. Finally, subtracting $\overline{X}\sum_i (Y_i - \overline{Y}) = 0$ from the top and $\overline{X}\sum_i (X_i - \overline{X}) = 0$ from the bottom (the toolkit trick) turns this into exactly the $\hat\beta_1$ formula.

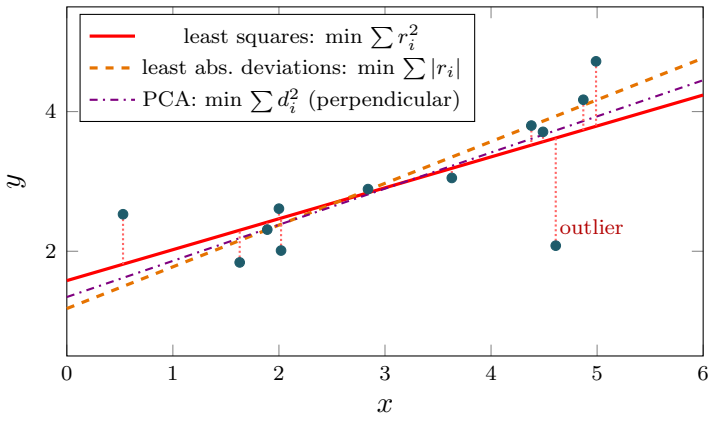


Figure 8: Made-up data with one outlier. Dotted red: the residuals r_i of the least squares line. Three criteria give three different lines. Because least squares *squares* the residuals, the outlier's big residual counts heavily, and it pulls the LS line down more than the least-absolute-deviations line. The perpendicular-distance (PCA) line treats x and y symmetrically. Minimizing SSR is the right choice when the data come from a linear regression model.

8.4 Two kinds of replicates

$\hat\beta_0, \hat\beta_1$ are computed from random data, so they have sample distributions. For regression there are two ways to make a replicate:

- **Conditional on X:** keep the same $X_1, \ldots, X_N$ and draw new $Y_i \mid X_i$. Example: an experiment where we choose the doses.

- **Not conditional on X:** draw new pairs (X_i, Y_i). Example: a survey, where a new sample brings new people with new X's.

We focus on the sample distribution **conditional on** $X_1, \dots, X_N$: the X's are fixed numbers and only the Y's are random. Write $S_{xx} = \sum_i (X_i - \overline{X})^2 \approx N\sigma_X^2$.

8.5 $\hat{\beta}_1$ is unbiased

Conditional on the X's, the denominator S_{xx} and each $(X_i - \overline{X})$ are constants, so linearity of $\mathbb{E}$ lets us move the expectation onto the Y's:

$$\mathbb{E}[\hat{\beta}_1 \mid X_1, \dots, X_N] = \frac{\sum_i (X_i - \overline{X})\, \mathbb{E}[(Y_i - \overline{Y}) \mid X_1, \dots, X_N]}{\sum_i (X_i - \overline{X})^2} \qquad \text{(linearity)}.$$

From the model, $\mathbb{E}[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$, and averaging over i, $\mathbb{E}[\overline{Y} \mid X_1, \dots, X_N] = \beta_0 + \beta_1 \overline{X}$. So

$$\mathbb{E}[(Y_i - \overline{Y}) \mid X_1, \dots, X_N] = \beta_0 + \beta_1 X_i - \beta_0 - \beta_1 \overline{X} = \beta_1 (X_i - \overline{X}),$$

and therefore

$$\mathbb{E}[\hat{\beta}_1 \mid X_1, \dots, X_N] = \frac{\sum_i (X_i - \overline{X})\, \beta_1 (X_i - \overline{X})}{\sum_i (X_i - \overline{X})^2} = \beta_1 \frac{\sum_i (X_i - \overline{X})^2}{\sum_i (X_i - \overline{X})^2} = \beta_1.$$

Across replicates with the same X's, $\hat{\beta}_1$ is right on average. (A similar argument shows $\mathbb{E}[\hat{\beta}_0 \mid X] = \beta_0$.) The key ingredient: the noise ϵ_i is independent of *all* the predictors. Section 9 shows what goes wrong without it.

8.6 The sample distribution of $\hat{\beta}_1$

Variance (optional derivation; the notes state it without proof). Because $\sum_i (X_i - \overline{X})\overline{Y} = \overline{Y} \cdot 0 = 0$, we can drop $\overline{Y}$ from the numerator: $\hat{\beta}_1 = \frac{1}{S_{xx}} \sum_i (X_i - \overline{X}) Y_i$. Given the X's, this is a weighted sum of independent Y_i's, each with variance σ_ϵ^2, so

$$\operatorname{var}(\hat{\beta}_1 \mid X) = \frac{1}{S_{xx}^2} \sum_i (X_i - \overline{X})^2 \sigma_\epsilon^2 = \frac{\sigma_\epsilon^2 S_{xx}}{S_{xx}^2} = \frac{\sigma_\epsilon^2}{S_{xx}}.$$

A weighted sum of Normals is Normal, and $S_{xx} \approx N\sigma_X^2$, so:

Sample distribution of the slope

$$\hat{\beta}_1 \approx \text{Normal}\left(\beta_1,\ \frac{\sigma_\epsilon^2}{N\sigma_X^2}\right), \qquad \text{se}(\hat{\beta}_1) \approx \frac{\sigma_\epsilon}{\sigma_X \sqrt{N}}.$$

The slope is pinned down better with **less noise** (small σ_ϵ), **more spread-out X's** (large σ_X), or **more data** (large N). Intuition for σ_X: if all X's are bunched together, a little noise can tilt the line a lot, like balancing a ruler on two close fingers.

In words: across repeated datasets the estimated slope is centered on the true slope, and it varies less when the noise is small, the X's are spread out, or there is more data.

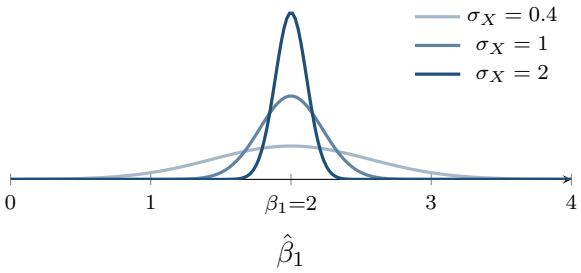


Figure 9: Normal$(\beta_1, \sigma_\epsilon^2/(N\sigma_X^2))$ with $N = 20$, $\beta_1 = 2$, $\sigma_\epsilon = 1$ (the 10/7 slide setting). Spreading the X's out by a factor of 5 shrinks se$(\hat{\beta}_1)$ by a factor of 5.

Back to the graders (10/7)

$N = 68$ exams, $N_E = 23$ graded by E, $\hat{\sigma}_\epsilon \approx 5$. Here $X \sim$ Bernoulli(p) with $p \approx 23/68 \approx 1/3$, so $\sigma_X^2 = p(1-p) \approx \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$ and $N\sigma_X^2 \approx 68 \cdot \frac{2}{9} = \frac{136}{9} \approx 15.1$. Then

$$\text{var}(\hat{\beta}_1) \approx \frac{\hat{\sigma}_\epsilon^2}{N\sigma_X^2} \approx \frac{25}{136/9} = \frac{225}{136} \approx 1.65, \qquad \text{se}(\hat{\beta}_1) \approx \sqrt{1.65} \approx 1.29.$$

Compare: back of the envelope, var $\approx 3/2$ (se ≈ 1.2); exact Python, var $= 1.84$ (se ≈ 1.36). All three roughly agree: the boxed formula gives the same answer as the difference of means.

8.7 Computing everything in Python

Using the covariance matrix:

```
C = np.cov(x, y)
beta1_hat = C[0, 1] / C[0, 0]          # cov(x, y) / var(x)
```

`C` is the 2×2 **sample covariance matrix** $\begin{pmatrix} \widehat{\text{var}}(X) & \widehat{\text{cov}}(X,Y) \\ \widehat{\text{cov}}(X,Y) & \widehat{\text{var}}(Y) \end{pmatrix}$ with $\widehat{\text{cov}}(X,Y) = \frac{1}{N-1}\sum_i (X_i - \overline{X})(Y_i - \overline{Y})$. Python counts rows and columns from 0, so `C[0, 1] / C[0, 0]` is $\widehat{\text{cov}}/\widehat{\text{var}}(X) = \hat{\beta}_1$ (the $\frac{1}{N-1}$'s cancel). Directly from the formulas:

```
beta1_hat = np.sum((x - np.mean(x)) * (y - np.mean(y))) \
            / np.sum((x - np.mean(x))**2)
beta0_hat = np.mean(y) - beta1_hat * np.mean(x)
r = y - (beta0_hat + beta1_hat * x)                  # residuals
sigma2_hat = np.sum(r**2) / (len(y) - 2)
R2 = 1 - np.sum(r**2) / np.sum((y - np.mean(y))**2)
se_beta1 = np.sqrt(sigma2_hat / np.sum((x - np.mean(x))**2))
T = beta1_hat / se_beta1
p_v = 2 * (1 - norm.cdf(np.abs(T)))                  # norm from scipy.stats
```

The last line is the p-value: $\mathbb{P}(|\hat{T}^*| > |\hat{T}|)$ is two tails, each $1 - \Phi(|\hat{T}|)$.

Example (notes): TV advertising and sales

The Advertising data has $N = 200$ markets, $X =$ TV budget (\$1000s), $Y =$ sales (1000s of units). Running the code above gives $\hat{\beta}_0 \approx 7.03$, $\hat{\beta}_1 \approx 0.048$, SSR ≈ 2103, SST ≈ 5417, $\hat{\sigma}_\epsilon \approx 3.26$, $R^2 \approx 0.61$.

Interpretation: each extra \$1000 of TV advertising goes with about 48 more units sold (0.048 thousand), and the TV budget explains about 61% of the variation in sales between markets. Check: $1 - 2103/5417 = 0.61$ and $\sqrt{2103/198} = 3.26$.

Model check by simulation: making "fake" datasets $\hat{\beta}_0 + \hat{\beta}_1 x + \text{Normal}(0, \hat{\sigma}_\epsilon^2)$ shows constant spread, while the real data fan out (residual SD about 1.8 for budgets under 100, about 4.4 over 200). So the constant-variance assumption is questionable here.

9 Autoregressive models and the Hurwicz bias (Unit 3.5)

9.1 Time series and the AR(1) model

A **time series** $Y_1, Y_2, \ldots, Y_N$ is a sequence of measurements of the same thing at successive times; the running example is $Y_t = \log(\text{bitcoin price on day } t)$. Goal: predict tomorrow's price, and the next day's...

Using time as the predictor does not really make sense: prices on different days are not produced independently, so the points (t, Y_t) and $(t+1, Y_{t+1})$ are not independent. Instead we capture the fact that today's price depends on yesterday's.

AR(1) model

An **autoregressive model** AR(1) is a regression whose predictor is the previous value of the response:

$$Y_t \mid Y_{t-1} \sim \text{Normal}(\beta_0 + \beta_1 Y_{t-1},\ \sigma_\epsilon^2), \qquad \text{i.e.} \qquad Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t,$$

where ϵ_t is independent of everything *before* time t (but not of the future). The regression data are the pairs

$$(X_1, Y_1), \ldots, (X_{N-1}, Y_{N-1}) \longleftrightarrow (Y_1, Y_2),\ (Y_2, Y_3), \ldots, (Y_{N-1}, Y_N).$$

In words: today's value is a straight-line function of yesterday's value plus fresh noise.

Once you know Y_{t-1}, the earlier values $Y_{t-2}, Y_{t-3}, \ldots$ tell you nothing more about Y_t. For bitcoin, least squares gives $\hat{\beta}_1 \approx 0.9996$ and $\hat{\sigma}_\epsilon \approx 0.016$ (with $\log_{10}$): today's log price is essentially yesterday's plus noise.

9.2 Stationary AR(1) processes

A series is **stationary** if its distribution does not change over time; in particular Y_t and Y_{t-1} have the same mean μ and the same standard deviation σ_Y. Three facts follow.

1. The slope is a correlation. The slope of any regression is $\text{cov}(X, Y)/\sigma_X^2$. Here $X = Y_{t-1}$ and

$Y = Y_t$ have the same SD, so

$$\beta_1 = \frac{\operatorname{cov}(Y_{t-1}, Y_t)}{\sigma_Y^2} = \frac{\operatorname{cov}(Y_{t-1}, Y_t)}{\sigma_Y\,\sigma_Y} = \rho,$$

the correlation between consecutive values. From now on we write ρ for β_1.

2. The mean. Take expectations of $Y_t = \beta_0 + \rho Y_{t-1} + \epsilon_t$, using $\mathbb{E}Y_t = \mathbb{E}Y_{t-1} = \mu$ and $\mathbb{E}\epsilon_t = 0$:

$$\mu = \beta_0 + \rho\mu \implies \beta_0 = (1-\rho)\mu, \qquad \text{so} \qquad Y_t = \mu(1-\rho) + \rho Y_{t-1} + \epsilon_t,$$

which can be rewritten as $Y_t - \mu = \rho(Y_{t-1} - \mu) + \epsilon_t$. When $|\rho| < 1$, each step shrinks the deviation from the mean by a factor ρ on average and then adds new noise: **regression to the mean**. Daughters of unusually tall mothers are tall, but on average less tall than their mothers.

3. The variance. Take variances; ϵ_t is independent of Y_{t-1}, so the variances add:

$$\sigma_Y^2 = \rho^2\sigma_Y^2 + \sigma_\epsilon^2 \implies \sigma_Y^2(1-\rho^2) = \sigma_\epsilon^2 \implies \sigma_Y^2 = \frac{\sigma_\epsilon^2}{1-\rho^2}.$$

The closer ρ is to 1, the more slowly the series returns to its mean, and the farther it wanders. Compare with Section 2: $1 - \rho^2 = \sigma_\epsilon^2/\sigma_Y^2$ is the same identity.

9.3 When $\rho \approx 1$: the random walk

If $\rho = 1$, the formula gives $\sigma_Y^2 = \infty$. How can that be? Take $\beta_0 = 0$, $Y_0 = 0$ and unroll the model:

$$Y_{t+1} = Y_t + \epsilon_{t+1} = Y_{t-1} + \epsilon_t + \epsilon_{t+1} = \cdots = \sum_{s=1}^{t+1} \epsilon_s.$$

Y_t is a sum of t independent Normal$(0, \sigma_\epsilon^2)$ steps, so

$$Y_t \sim \text{Normal}(0,\ \sigma_\epsilon^2\, t).$$

This is a **random walk**: the noise is never forgotten, and the variance grows forever instead of settling down. There is no stationary distribution. With $\hat\beta_1 \approx 0.9996$, the bitcoin log price is very close to a random walk.

9.4 Why least squares is biased here: the Hurwicz bias

Take $\mu = 0$ for simplicity and fit $Y_t = \rho Y_{t-1} + \epsilon_t$ by least squares with no intercept (then $\hat\rho = \sum Y_{t-1}Y_t / \sum Y_{t-1}^2$). Substitute the model into the numerator:

$$\begin{aligned}
\mathbb{E}[\hat\rho] = \mathbb{E}\left[\frac{\sum_t Y_{t-1}Y_t}{\sum_t Y_{t-1}^2}\right] &= \mathbb{E}\left[\frac{\sum_t Y_{t-1}(\rho Y_{t-1} + \epsilon_t)}{\sum_t Y_{t-1}^2}\right] && (Y_t = \rho Y_{t-1} + \epsilon_t) \\
&= \rho + \mathbb{E}\left[\frac{\sum_t Y_{t-1}\epsilon_t}{\sum_t Y_{t-1}^2}\right] \underset{\text{Hurwicz (1950)}}{\approx} \rho - \frac{2\rho}{N}. && (\text{split the fraction})
\end{aligned}$$

Why the second term is not zero. In ordinary regression the noise ϵ_i is independent of *all* of the predictors, so conditional on the X's the second term has mean 0 and $\hat\beta_1$ is unbiased (that was the proof in Section 8). In a time series the predictors are $Y_1, \ldots, Y_{N-1}$ themselves. Although ϵ_t is independent of the *past* value Y_{t-1}, it is part of Y_t, Y_{t+1}, and every later value, all of which appear

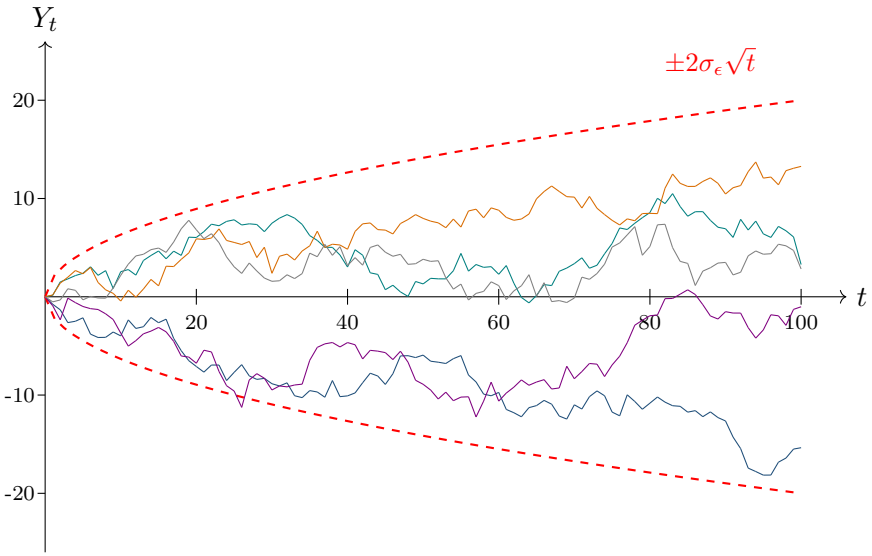


Figure 10: Five simulated random walks with $\sigma_\epsilon = 1$. The envelope $\pm 2\sigma_\epsilon\sqrt{t}$ (about 95% of walks at each t) keeps widening.

in the denominator. The noise is correlated with the predictors that come after it, so the argument breaks down.

Hurwicz bias

For a finite series, least squares underestimates how persistent an AR(1) process is: $\mathbb{E}[\hat{\rho}] - \rho \approx -2\rho/N$. The estimator is biased (pulled toward 0) but consistent (the bias vanishes as $N \to \infty$). You do not need to memorize the formula; know *why* the bias exists.

In words: with a short series, the fitted $\hat{\rho}$ comes out too small on average, by about $2\rho/N$.

Example (notes): seeing the bias by simulation

```
rng = np.random.default_rng(42)
beta1 = 0.6

def fit_beta1_hat(N):
    eps = rng.normal(size=N)
    Y = np.empty(N)
    Y[0] = eps[0]
    for t in range(1, N):
        Y[t] = beta1 * Y[t - 1] + eps[t]
    y_lag, y = Y[:-1], Y[1:]
    return np.sum(y_lag * y) / np.sum(y_lag**2)

for N in [10, 50, 200]:
    beta1_hats = [fit_beta1_hat(N) for _ in range(20_000)]
    print(f"N={N}: average beta1_hat = {np.mean(beta1_hats):.3f}")
```

prints averages 0.506, 0.577, 0.595 for $N = 10, 50, 200$. Every one falls short of the true 0.6, and the gap shrinks with N. The formula predicts biases $-2(0.6)/50 = -0.024$ and $-2(0.6)/200 = -0.006$, matching the simulation; at $N = 10$ the match is worse, because the formula is only a large-N

approximation.

9.5 How fast do correlations decay? Autocorrelation

The **autocorrelation at lag** k is $\rho_k = \operatorname{corr}(Y_t, Y_{t+k})$. For a stationary AR(1) with mean 0, $Y_{t+1} = \rho Y_t + \epsilon_{t+1}$, apply the model repeatedly:

$$\begin{aligned} Y_{t+1} &= \rho Y_t + \epsilon_{t+1} \\ Y_{t+2} &= \rho Y_{t+1} + \epsilon_{t+2} = \rho(\rho Y_t + \epsilon_{t+1}) + \epsilon_{t+2} = \rho^2 Y_t + \rho\,\epsilon_{t+1} + \epsilon_{t+2} \\ &\vdots \\ Y_{t+k} &= \rho^k Y_t + \underbrace{\rho^{k-1}\epsilon_{t+1} + \cdots + \rho\,\epsilon_{t+k-1} + \epsilon_{t+k}}_{\text{future noise, independent of } Y_t}. \end{aligned}$$

By bilinearity, $\operatorname{cov}(Y_t, Y_{t+k}) = \rho^k \operatorname{cov}(Y_t, Y_t) + 0 = \rho^k \sigma_Y^2$, and since $\operatorname{var}(Y_{t+k}) = \operatorname{var}(Y_t) = \sigma_Y^2$:

Autocorrelation of an AR(1)

$$\rho_k = \frac{\operatorname{cov}(Y_t, Y_{t+k})}{\sigma_Y \sigma_Y} = \rho^k.$$

The correlation decays *exponentially* in the lag. After $k = 10$ steps: $0.5^{10} \approx 0.001$, $0.9^{10} \approx 0.35$, $0.99^{10} \approx 0.90$.

In words: values k steps apart have correlation ρ to the power k, so the series' memory fades by a factor of ρ each step.

From data we use the **sample autocorrelation** $\hat\rho_k = \dfrac{\sum_t (Y_t - \overline{Y})(Y_{t+k} - \overline{Y})}{\sum_t (Y_t - \overline{Y})^2}$; a plot of $\hat\rho_k$ against k is the **autocorrelation function**. For simulated AR(1) series $\hat\rho_k$ follows ρ^k. The bitcoin log price stays correlated with itself for a very long time ($\rho \approx 1$), while its daily changes $Y_t - Y_{t-1}$ are almost uncorrelated: how the price moved today says very little about how it will move tomorrow.

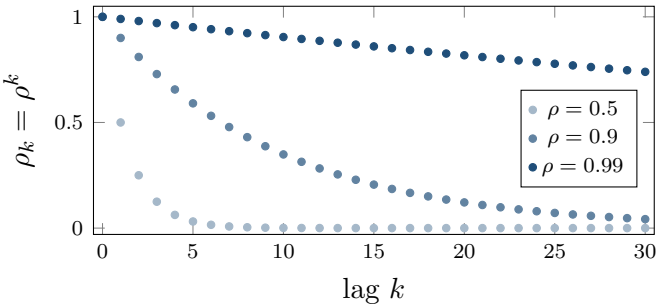


Figure 11: Autocorrelation functions of AR(1) processes. With $\rho = 0.5$ the series forgets its past within a few steps; with $\rho = 0.99$ it remembers for a very long time.

9.6 Forecasting: random walk vs. linear extrapolation

On the 10/9 slides two models are fit to 180 days of bitcoin log price and asked to forecast 90 days ahead:

- **Linear in time:** fit $Y_t = \beta_0 + \beta_1 t + \epsilon_t$ and extend the line. The 95% band includes both the noise and the uncertainty in the fitted line.
- **Random walk:** forecast $Y_{t+h} \approx Y_t$ with 95% band $\pm 1.96\,\hat{\sigma}_\epsilon \sqrt{h}$. Why: unrolling, $Y_{t+h} = Y_t + \epsilon_{t+1} + \cdots + \epsilon_{t+h}$, a sum of h independent noises with total variance $h\sigma_\epsilon^2$.

The linear trend confidently extrapolates whatever trend the last 180 days happened to have, and it was badly wrong when the market turned. The random-walk band widens like $\sqrt{h}$ and honestly reflects that the future is uncertain.

9.7 Prices vs. log prices (optional)

So far $Y_t = \log P_t$. If the log price is (close to) a random walk, $Y_{t+1} = Y_t + \epsilon_{t+1}$, then exponentiating,

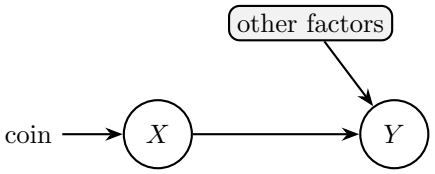
$$P_{t+1} = e^{Y_{t+1}} = e^{Y_t} e^{\epsilon_{t+1}} = P_t\, e^{\epsilon_{t+1}} \approx P_t(1 + \epsilon_{t+1}).$$

The last step is the tangent-line approximation $e^x \approx 1 + x$ for small x (the derivative of e^x at 0 is 1). So each day the price is *multiplied* by a random factor: a **multiplicative process**. Then $P_{t+1} - P_t \approx P_t\,\epsilon_{t+1}$: the daily change is proportional to the current price. The relative volatility is constant ($\hat{\sigma}_\epsilon \approx 0.04$ in natural log, about 4% a day; this is the 0.016 above times $\ln 10$), but in dollars the swings grow with the price: a typical daily move was about \$3 in 2015 and about \$900 in the last year.

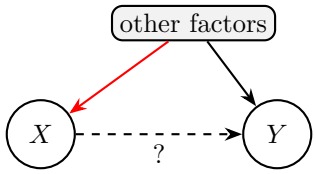
10 Study design and multiple predictors (Unit 4.1–4.2)

10.1 Two ways to collect data

Does a treatment ($X = 1$) change an outcome Y, compared with no treatment ($X = 0$)?



Randomized controlled trial
X independent of all other factors



Cohort study (observational)
X may be tied to other factors

- **Randomized controlled trial (RCT):** we assign X by a coin flip, so X is independent of every other factor, measured or not. Any difference in Y between the groups must come from X.
- **Cohort study:** we follow people and observe who gets X. Then X may be correlated with other factors that also affect Y (the red arrow), and we cannot tell what causes what. Such a factor is a **confounder**.

For most questions in the social sciences an RCT is impossible. Multiple regression is the main tool for *controlling* for the confounders we can measure.

10.2 Example: children's test scores

$N = 434$ children. Y: the child's test score; $X_{hs} = 1$ if the mother finished high school (0 if not); X_{iq}: the mother's IQ. (IQ is an imperfect measure of cognitive ability: one test score that also depends on the test, the conditions, and the person's background.)

Standard deviations: $\sigma_Y = 20.4$, $\sigma_{HS} = 0.41$, $\sigma_{IQ} = 15$. We write β' (with a prime) for a *single-predictor* coefficient, to distinguish it from the coefficients β of the two-predictor model.

- Y on X_{hs}: $\hat{\beta}'_{hs} = 11.8$ (the difference in average score between the two groups), $\hat{\rho} = \hat{\beta}'_{hs}\sigma_{HS}/\sigma_Y \approx 0.24$.
- Y on X_{iq}: $\hat{\beta}'_{iq} = 0.61$, $\hat{\rho} = \hat{\beta}'_{iq}\sigma_{IQ}/\sigma_Y \approx 0.45$.
- Mothers who finished high school have higher average IQ: 102.2 vs. 91.9.

10.3 What we computed vs. what we want

- **Computed from the single-predictor regression:** the difference in average score between the groups, $\beta'_{hs} = \mathbb{E}[Y \mid X_{hs} = 1] - \mathbb{E}[Y \mid X_{hs} = 0]$ ($\hat{\beta}'_{hs} = 11.8$). Part of this is really an IQ difference.
- **What we want:** the difference *with IQ held fixed*, $\beta_{hs} = \mathbb{E}[Y \mid X_{hs} = 1, X_{iq} = x] - \mathbb{E}[Y \mid X_{hs} = 0, X_{iq} = x]$.
- To compare children whose mothers have the same IQ, put both variables in the model:

$$Y = \beta_0 + \beta_{hs}X_{hs} + \beta_{iq}X_{iq} + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

Fitting it in Python with statsmodels:

```
import statsmodels.api as sm

X = sm.add_constant(np.column_stack([mom_hs, mom_iq]))
results = sm.OLS(kid_score, X).fit()
print(results.summary())
```

```
Dep. Variable:                      y   R-squared:                       0.214
No. Observations:                 434   Adj. R-squared:                  0.210
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         25.7315      5.875      4.380      0.000      14.184      37.279
mom_hs         5.9501      2.212      2.690      0.007       1.603      10.297
mom_iq         0.5639      0.061      9.309      0.000       0.445       0.683
```

The `coef` column has the least squares estimates $\hat{\beta}_0 = 25.7$, $\hat{\beta}_{hs} = 5.95$, $\hat{\beta}_{iq} = 0.56$ (`add_constant` adds the intercept). `std err` is $\text{se}(\hat{\beta}_k)$, `t` is $\hat{T} = \hat{\beta}_k/\text{se}$, `P>|t|` is the p-value, and the last two columns are the 95% CI (all t-based). Controlling for IQ, finishing high school goes with 5.95 more points, about half of 11.8.

10.4 From two predictors to one

Pick a mom without high school (IQ $= a$) and one with (IQ $= b$). The model says their children's expected scores differ by $\beta_{hs} + \beta_{iq}(b - a)$: the schooling effect plus the IQ difference times the IQ slope.

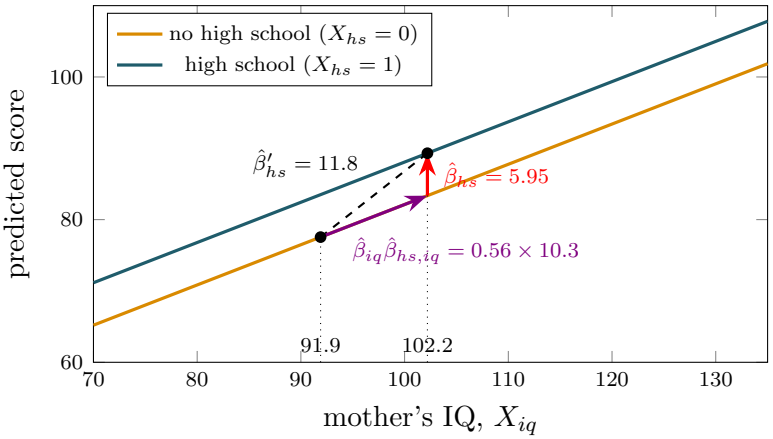


Figure 12: The fitted two-predictor model is two parallel lines (slope $\hat{\beta}_{iq}$, gap $\hat{\beta}_{hs}$). The dots are the two groups' averages. Going from the no-high-school average to the high-school average, the total rise $\hat{\beta}'_{hs}$ is a move along the line (purple: the IQ difference times the IQ slope) plus a jump between lines (red: the schooling effect with IQ fixed).

Averaging over all moms in each group replaces a and b by the group-average IQs:

$$\underbrace{\mathbb{E}[Y \mid X_{hs}{=}1] - \mathbb{E}[Y \mid X_{hs}{=}0]}_{\beta'_{hs}} = \beta_{hs} + \beta_{iq} \underbrace{\left(\mathbb{E}[X_{iq} \mid X_{hs}{=}1] - \mathbb{E}[X_{iq} \mid X_{hs}{=}0]\right)}_{\beta_{hs,iq}}.$$

Here $\beta_{hs,iq}$ is the slope when IQ is regressed on schooling. With the numbers: $11.8 = 5.95 + 0.56 \times 10.3$ (the right side is 11.76). We say β_{hs} is the effect of schooling **controlling for** IQ (and vice versa). Section 11 proves this decomposition in general.

10.5 Which predictor matters more?

$\hat{\beta}_{hs} = 5.95$ is ten times $\hat{\beta}_{iq} = 0.56$, but they have different units (points for finishing high school vs. points per IQ point), so comparing them directly is meaningless. Instead compare the change in average score for a *typical* change in each predictor, one standard deviation:

Scaled coefficients

To compare predictors with different units, compare $\hat{\beta}_k \, \sigma_{X_k}$ (the effect of a one-SD change in X_k).

In words: multiply each coefficient by its predictor's standard deviation to get "points of Y per typical change in X_k"; these are comparable across predictors.

```
b = results.params
print(b[1] * np.std(mom_hs))   # high school -> 2.441
print(b[2] * np.std(mom_iq))   # IQ          -> 8.449
```

A one-SD difference in IQ (15 points) goes with about 8.4 more points; one SD in schooling (0.41) with only about 2.4. IQ matters more. (The same scaling applies to the CIs: the high-school CI $[1.6, 10.3]$ becomes about $[0.66, 4.2]$.)

10.6 R^2 with multiple predictors

R^2 is defined exactly as before, $R^2 = 1 - \text{SSR}/\text{SST}$, where now the fitted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1,i} + \cdots + \hat{\beta}_K X_{K,i}$ uses all K predictors ($X_{k,i}$ is the value of predictor k for data point i), and $\text{SSR} = \sum_i (Y_i - \hat{Y}_i)^2$. It is the fraction of the variation in Y explained by all the predictors together: $R^2 \approx 1 - \text{var}(Y \mid X_1, \ldots, X_K)/\text{var}(Y)$.

predictors	R^2
high school only	0.06
IQ only	0.20
high school and IQ	0.21

- Adding a predictor never decreases R^2 (proved in Section 11).
- The R^2's do not add up ($0.06 + 0.20 > 0.21$): mothers who finished high school also tend to have higher IQs, so much of what schooling explains is already explained by IQ.
- About 79% of the variation is *not* explained: scores have SD 20 and the residuals still have $\hat{\sigma}_\epsilon \approx 18$. The model says little about any individual child's score.

Question (10/12): midterm scores, difficulty, and QSS

$N = 64$ students. y = midterm score (fraction of points), `difficult` = 1 if the student rated difficulty 4 (the highest) in the week 1 survey, `QSS` = 1 for QSS majors. Fit: $\hat{\beta}_0 = 0.8705$, $\hat{\beta}_{diff} = -0.1002$ (se 0.040), $\hat{\beta}_{QSS} = -0.1340$ (se 0.040); `np.std(difficult)` = 0.498, `np.std(QSS)` = 0.499. The joint proportions are:

	`difficult`= 0	`difficult`= 1	total
`QSS`= 0	0.281	0.188	0.469
`QSS`= 1	0.266	0.266	0.531
total	0.547	0.453	1

Which predictor is more important? Both are 0/1 with SD ≈ 0.5, so scaling barely changes anything: $\hat{\beta}_{diff}\sigma_{diff} = -0.100 \times 0.498 = -0.050$, $\hat{\beta}_{QSS}\sigma_{QSS} = -0.134 \times 0.499 = -0.067$. QSS is (slightly) more important; here comparing the raw coefficients would have given the same answer.
Single-predictor coefficient of difficulty. Use $\beta'_{diff} = \beta_{diff} + \beta_{QSS}\,\beta_{diff,QSS}$, where $\beta_{diff,QSS}$ is the slope of QSS on difficulty, a difference of conditional means read off the table (divide a cell by its column total):

$$\hat{\beta}_{diff,QSS} = \mathbb{E}[X_{QSS} \mid X_{diff}{=}1] - \mathbb{E}[X_{QSS} \mid X_{diff}{=}0] = \frac{0.266}{0.453} - \frac{0.266}{0.547} = 0.587 - 0.486 = 0.101,$$

so $\hat{\beta}'_{diff} = -0.100 + (-0.134)(0.101) = -0.114$. Fitting y on `difficult` alone gives -0.1137, as predicted.

11 Omitted-variable bias and Simpson's paradox

11.1 The theorem

Theorem

Suppose $Y = \beta_0+\beta_1X_1+\beta_2X_2+\epsilon$, where ϵ is uncorrelated with X_1 and X_2, and let $\rho^2 = 1-\sigma_\epsilon^2/\sigma_Y^2$.

(1) **Omitted-variable bias:** if X_2 is left out, the slope of the regression of Y on X_1 is

$$\beta_1' = \beta_1 + \beta_2\,\beta_{1,2}, \qquad \beta_{1,2} = \frac{\operatorname{cov}(X_1, X_2)}{\operatorname{var}(X_1)} \quad \text{(the slope of } X_2 \text{ regressed on } X_1\text{)}.$$

(2) Adding a predictor never decreases ρ^2.

The same argument with sample covariances shows R^2 never decreases. Both results extend to K predictors.

In words: (1) leaving X_2 out makes X_1's slope absorb X_2's effect, scaled by how much X_2 rises with X_1; (2) adding a predictor can never explain less of Y's variance.

Reading (1). Leaving X_2 out does not make its effect disappear; it gets credited to X_1, in proportion to how much X_2 moves with X_1. The bias $\beta_2\beta_{1,2}$ is zero only if X_2 has no effect ($\beta_2 = 0$) or is uncorrelated with X_1 ($\beta_{1,2} = 0$). The second case is exactly what randomization guarantees in an RCT.

Proof of (1). For simplicity assume X_1, X_2 have mean zero and all intercepts are zero. The single-predictor slope is $\beta_1' = \operatorname{cov}(X_1, Y)/\operatorname{var}(X_1)$. Using bilinearity of covariance and $\operatorname{cov}(X_1, \epsilon) = 0$:

$$\begin{aligned}
\operatorname{cov}(X_1, Y) &= \operatorname{cov}(X_1,\ \beta_1X_1 + \beta_2X_2 + \epsilon) && \text{(substitute the model)}\\
&= \beta_1 \operatorname{cov}(X_1, X_1) + \beta_2 \operatorname{cov}(X_1, X_2) + \operatorname{cov}(X_1, \epsilon) && \text{(bilinearity)}\\
&= \beta_1 \operatorname{var}(X_1) + \beta_2 \operatorname{cov}(X_1, X_2). && (\operatorname{cov}(X_1, X_1) = \operatorname{var} X_1,\ \operatorname{cov}(X_1, \epsilon) = 0)
\end{aligned}$$

Dividing by $\operatorname{var}(X_1)$: $\beta_1' = \beta_1 + \beta_2\dfrac{\operatorname{cov}(X_1, X_2)}{\operatorname{var}(X_1)} = \beta_1 + \beta_2\beta_{1,2}$. □

Proof of (2). The coefficients of determination of the two- and one-predictor models are

$$\rho^2 = 1 - \frac{\operatorname{var}(Y \mid X_1, X_2)}{\sigma_Y^2}, \qquad \rho'^2 = 1 - \frac{\operatorname{var}(Y \mid X_1)}{\sigma_Y^2},$$

so we just need $\operatorname{var}(Y \mid X_1) \geq \operatorname{var}(Y \mid X_1, X_2)$. Knowing both predictors, the only randomness left is the noise: $\operatorname{var}(Y \mid X_1, X_2) = \sigma_\epsilon^2$. Knowing only X_1, both X_2 and ϵ are still random; β_1X_1 is then a constant, and since ϵ is independent of X_1, X_2 the variances add:

$$\operatorname{var}(Y \mid X_1) = \operatorname{var}(\beta_2X_2 + \epsilon \mid X_1) = \beta_2^2 \operatorname{var}(X_2 \mid X_1) + \sigma_\epsilon^2 \ \geq\ \sigma_\epsilon^2. \qquad \square$$

In words: more information can only leave less unexplained variance (or the same, if the new predictor is useless).

Picture. With two continuous predictors the model is a plane over the (x_1, x_2) plane. Moving one unit in x_1 with x_2 fixed raises y by β_1. But in the data, when x_1 goes up by one, x_2 typically goes up by $\beta_{1,2}$ too (the points cluster along the line $x_2 = \beta_{1,2}x_1$), adding another $\beta_2\beta_{1,2}$. The single-predictor slope β_1' is the slope of the plane *along that line*.

Question (10/14): poverty, single-parent households, and violent crime

50 US states (2009). Y: violent crimes per 100,000; X_{pov}: % below the poverty line; X_{single}: % of family households headed by a single parent. Three fits:

model	coefficient on poverty	coefficient on single	R^2
poverty only	15.97 (se 7.19, $p = 0.031$)	—	0.093
single only	—	29.99 (se 4.92, $p < 0.001$)	0.436
both	-1.29 (se 6.57, $p = 0.845$)	30.54 (se 5.70, $p < 0.001$)	0.437

Interpretation. Alone, poverty looks associated with crime (16 more crimes per 100,000 per percentage point). Controlling for single-parent households, that association vanishes (the coefficient is near zero with a CI containing 0), while the single-parent coefficient barely changes. States with more poverty have more single-parent households; the poverty-only regression was picking up the single-parent effect. (This is an observational study, so these are associations, not causes.)
Find $\beta_{pov,single}$**.** Use OVB with $X_1 =$ poverty, $X_2 =$ single: $\beta'_{pov} = \beta_{pov} + \beta_{single}\beta_{pov,single}$, so

$$15.97 = -1.29 + 30.54\,\hat{\beta}_{pov,single} \implies \hat{\beta}_{pov,single} = \frac{15.97 + 1.29}{30.54} = 0.565.$$

The other way around, $\beta'_{single} = \beta_{single} + \beta_{pov}\beta_{single,pov}$: $29.99 = 30.54 - 1.29\,\hat{\beta}_{single,pov}$, so $\hat{\beta}_{single,pov} = 0.55/1.29 = 0.425$.
The correlation between the predictors. Multiply the two slopes:

$$\beta_{pov,single}\,\beta_{single,pov} = \frac{\text{cov}}{\text{var}(X_{pov})} \cdot \frac{\text{cov}}{\text{var}(X_{single})} = \left(\frac{\text{cov}}{\sigma_{pov}\sigma_{single}}\right)^2 = \rho^2,$$

so $\hat{\rho} = \sqrt{0.565 \times 0.425} = \sqrt{0.240} = 0.49$ (positive, since both slopes are positive).

11.2 Simpson's paradox

	1995	1996	both years
Derek Jeter	12/48 = .250	183/582 = .314	195/630 = .310
David Justice	104/411 = .253	45/140 = .321	149/551 = .270

Batting average = hits / at bats. Justice has the higher average *in each year*, but Jeter has the higher average *overall*. This reversal is **Simpson's paradox**.

Question (10/14): explain it with omitted-variable bias

Let $Y = 1$ for a hit (0 otherwise), $X_{jeter} = 1$ for Jeter (0 for Justice), $X_{year} = 1$ for 1996 (0 for 1995), and consider $\mathbb{E}[Y \mid X_{jeter}, X_{year}] = \beta_0 + \beta_{jeter}X_{jeter} + \beta_{year}X_{year}$. Matching the four cells of the table:

$$\begin{aligned}
\beta_0 &\approx \mathbb{E}[Y \mid X_{jeter}{=}0, X_{year}{=}0] = 104/411 = .253,\\
\beta_0 + \beta_{jeter} &\approx \mathbb{E}[Y \mid X_{jeter}{=}1, X_{year}{=}0] = 12/48 = .250,\\
\beta_0 + \beta_{year} &\approx \mathbb{E}[Y \mid X_{jeter}{=}0, X_{year}{=}1] = 45/140 = .321,\\
\beta_0 + \beta_{jeter} + \beta_{year} &\approx \mathbb{E}[Y \mid X_{jeter}{=}1, X_{year}{=}1] = 183/582 = .314.
\end{aligned}$$

Only approximately: four averages cannot be matched exactly by three β's (that needs an *interaction* term $\beta_{int}X_{jeter}X_{year}$, coming soon). Reading off differences, $\hat{\beta}_{jeter} \approx -0.006$ (Justice is slightly better in both years) and $\hat{\beta}_{year} \approx 0.067$ (everyone hit better in 1996).
The omitted variable is the year. The table also gives how the year depends on the player: $\mathbb{E}[X_{year} \mid X_{jeter}{=}1] = 582/630 = .924$ and $\mathbb{E}[X_{year} \mid X_{jeter}{=}0] = 140/551 = .254$, so $\hat{\beta}_{jeter,year} = .924 - .254 = .670$: almost all of Jeter's at bats came in the good year. By OVB,

$$\hat{\beta}'_{jeter} = \hat{\beta}_{jeter} + \hat{\beta}_{year}\,\hat{\beta}_{jeter,year} = -0.006 + 0.067 \times 0.670 = 0.039,$$

matching the overall difference $.310 - .270 = .040$. A small negative direct effect is swamped by a large positive omitted-variable term.

Other examples of Simpson's paradox (each is OVB with a confounder):

- **UC Berkeley graduate admissions, 1973:** men admitted at a higher rate overall (44% vs. 35%), but in most departments women were admitted at a similar or higher rate. Omitted variable: department; women applied more often to departments that admitted few applicants.
- **Kidney stones:** open surgery succeeded more often than a newer procedure for small stones (93% vs. 87%) and for large stones (73% vs. 69%), but less often overall (78% vs. 83%). Omitted variable: stone size; open surgery was used mostly on the harder, large stones.
- **Smoking and survival (Whickham, UK):** over 20 years a higher fraction of smokers than non-smokers survived (76% vs. 69%), but within most age groups smokers died at higher rates. Omitted variable: age; the smokers were younger.
- **COVID-19, early 2020:** the case fatality rate was higher in Italy than in China overall, but lower in every age group. Omitted variable: age; Italy's cases were older.

Question (10/14): regressing an AR(1) on two lags

For a stationary AR(1) with coefficient ρ, fit $Y_t = \beta_0 + \beta_{lag1}Y_{t-1} + \beta_{lag2}Y_{t-2} + \epsilon$. What are β_{lag1} and β_{lag2}? What if we regress Y_t on Y_{t-2} alone?
Two lags: the AR(1) model says that once Y_{t-1} is known, Y_{t-2} tells us nothing more about Y_t, so $\beta_{lag1} = \rho$ and $\beta_{lag2} = 0$.
Y_{t-2} **alone:** $\beta'_{lag2} = \mathrm{cov}(Y_{t-2}, Y_t)/\sigma_Y^2 = \rho_2 = \rho^2$ (autocorrelation at lag 2). OVB gives the same answer: leaving out Y_{t-1}, $\beta'_{lag2} = \beta_{lag2} + \beta_{lag1} \cdot$ (slope of Y_{t-1} on Y_{t-2}) $= 0 + \rho \cdot \rho = \rho^2$. Yesterday's effect gets credited to the day before yesterday.

Math 50 Lectures 8–13 · Formula sheet

Algebra

$\text{var}(aX+b) = a^2 \text{var}\, X$; independent:
$\text{var}(X \pm Y) = \text{var}\, X + \text{var}\, Y$.
$\text{cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y$;
$\text{cov}(X, aY+bZ+c) = a\,\text{cov}(X,Y) + b\,\text{cov}(X,Z)$.
iid mean: $\mathbb{E}\overline{Y} = \mu$, $\text{var}\,\overline{Y} = \sigma^2/N$. $\sum_i (Y_i - \overline{Y}) = 0$.
Binomial(N,q): mean Nq, var $Nq(1-q)$.

Correlation (2.4)

$\rho = \dfrac{\beta_1 \sigma_X}{\sigma_Y} = \dfrac{\text{cov}(X,Y)}{\sigma_X \sigma_Y}$ = slope of Z_Y on Z_X.
$\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2$; $1 - \rho^2 = \dfrac{\sigma_\epsilon^2}{\sigma_Y^2} = \dfrac{\text{var}(Y \mid X)}{\text{var}\, Y}$.
ρ^2 = fraction of variance explained, in $[0,1]$.

LLN / CLT (3.1)

$S_N/N \to \mathbb{E}X_i$. $S_N \approx \text{Normal}(N\mu, N\sigma^2)$,
$\overline{Y} \approx \text{Normal}(\mu, \sigma^2/N)$ (iid, finite variance).

Estimators (3.2)

$\hat\theta$ random, θ fixed. Sample distribution = distribution of $\hat\theta$ over replicates; $\text{se}(\hat\theta)$ = its SD.
Normal mean: $\hat\mu \sim \text{Normal}(\mu, \sigma^2/N)$, se $= \sigma/\sqrt{N}$ ($\hat\sigma/\sqrt{N}$ in practice).
Bernoulli: $\hat q \approx \text{Normal}(q, q(1-q)/N)$.

MSE $= \mathbb{E}[(\hat\theta - \theta)^2] = \text{se}^2 + \text{Bias}^2$, Bias $= \mathbb{E}\hat\theta - \theta$.
Unbiased: Bias $= 0$ for all N. Consistent: $\mathbb{E}\hat\theta \to \theta$, se $\to 0$.

Laplace $\hat q_L = \frac{S+1}{N+2}$: bias $\frac{1-2q}{N+2}$, var $\frac{Nq(1-q)}{(N+2)^2}$.
$\hat\sigma_0^2 = \frac{1}{N}\sum (Y_i - \overline{Y})^2$: $\mathbb{E} = \frac{N-1}{N}\sigma^2$. Bessel:
$\hat\sigma^2 = \frac{1}{N-1}\sum (Y_i - \overline{Y})^2$ unbiased.
Key identity: $\sum (Y_i - \overline{Y})^2 = \sum (Y_i - \mu)^2 - N(\overline{Y} - \mu)^2$.

CIs and tests (3.3, 3.6)

95% CI: $\hat\theta \pm 1.96\,\text{se}(\hat\theta)$ (t: 2.78/2.26/2.05 for N=5/10/30).
$\hat T = \hat\beta_1/\text{se}(\hat\beta_1)$; $p_v = \mathbb{P}(|\hat T^*| > |\hat T|) = 2(1 - \Phi(|\hat T|))$.

CI meaning: over replicates, 95% of CIs contain θ (not "95% chance for this interval").
CI excludes 0 $\iff$ $|\hat T| > 1.96$ $\iff$ $p_v < 0.05$.
Width $\propto 1/\sqrt{N}$: $N \geq (1.96\sigma/\text{half-width})^2$.
Binary X, $N/2$ per group: $\hat\beta_1 \sim \text{Normal}(\beta_1, 4\sigma_\epsilon^2/N)$;
unequal groups: var $= \sigma_\epsilon^2(\frac{1}{N_0} + \frac{1}{N_1})$.
Density of Z near 0 ≈ 0.4: $\mathbb{P}(|Z| < a) \approx 0.8a$ for small a.
$\hat T^2 = (N-2)\dfrac{R^2}{1-R^2}$: small R^2 & tiny p_v possible (big N); big R^2 & big p_v impossible.

Least squares (3.4)

$\hat\beta_1 = \dfrac{\sum (X_i - \overline{X})(Y_i - \overline{Y})}{\sum (X_i - \overline{X})^2}$, $\hat\beta_0 = \overline{Y} - \hat\beta_1 \overline{X}$ (minimize SSR).
$r_i = Y_i - \hat\beta_0 - \hat\beta_1 X_i$, SSR $= \sum r_i^2$, SST $= \sum (Y_i - \overline{Y})^2$.
$\hat\sigma_\epsilon^2 = \text{SSR}/(N-2)$, $R^2 = 1 - \text{SSR/SST}$.

Conditional on X: $\mathbb{E}\hat\beta_1 = \beta_1$;
$\hat\beta_1 \approx \text{Normal}\Big(\beta_1, \dfrac{\sigma_\epsilon^2}{N\sigma_X^2}\Big)$, se $\approx \dfrac{\sigma_\epsilon}{\sigma_X\sqrt{N}}$.

`np.cov(x,y)[0,1]/np.cov(x,y)[0,0]` $= \hat\beta_1$.
`2*(1-norm.cdf(np.abs(T)))` $= p_v$.

AR(1) (3.5)

$Y_t = \beta_0 + \rho Y_{t-1} + \epsilon_t$; stationary: $\beta_1 = \rho$, $\beta_0 = (1-\rho)\mu$,
$Y_t - \mu = \rho(Y_{t-1} - \mu) + \epsilon_t$, $\sigma_Y^2 = \dfrac{\sigma_\epsilon^2}{1-\rho^2}$.
Autocorrelation $\rho_k = \rho^k$ (exponential decay).
Random walk ($\rho = 1$): $Y_t \sim \text{Normal}(0, \sigma_\epsilon^2 t)$; forecast $Y_t \pm 1.96\hat\sigma_\epsilon\sqrt{h}$.
Hurwicz: $\mathbb{E}\hat\rho \approx \rho - 2\rho/N$ (biased down, consistent) because ϵ_t is part of later predictors.
Log random walk $\Rightarrow P_{t+1} \approx P_t(1 + \epsilon_{t+1})$ (multiplicative).

Multiple predictors (4.1–4.2)

RCT: X independent of confounders. Cohort: not.
$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$: β_1 = effect of X_1 with X_2 held fixed.
Compare predictors with $\hat\beta_k \sigma_{X_k}$.
R^2 never decreases with more predictors; R^2's do not add.

OVB: $\beta_1' = \beta_1 + \beta_2\beta_{1,2}$, $\beta_{1,2} = \dfrac{\text{cov}(X_1, X_2)}{\text{var}\, X_1}$
(slope of X_2 on X_1; binary X_1: difference of means of X_2).

$\beta_{1,2}\,\beta_{2,1} = \rho_{12}^2$.
Simpson's paradox = OVB with a confounder strong enough to flip the sign.
AR(1) on two lags: $\beta_{lag1} = \rho$, $\beta_{lag2} = 0$; on lag 2 alone: ρ^2.

Worked numbers

Graders: se ≈ 1.2–1.36, $\hat T \approx 0.4$, $p_v \approx 0.7$.
Test scores: $11.8 = 5.95 + 0.56 \times 10.3$; scaled 2.4 vs. 8.4.
Jeter: $-0.006 + 0.067 \times 0.670 = 0.039$.
Crime: $\hat\beta_{pov,single} = 0.565$, $\hat\rho = 0.49$.

Symbol key (full glossary: p. 4)

$\hat\theta$ estimate from data $\overline{Y}$ sample average β' coefficient with the predictor alone
$Y \mid X$ "Y given X" $\sim$ "is distributed as" $\sum_i$ add over data points
$\approx$ approximately $\Rightarrow$ implies $\mathbb{E}$ expected value (true average)
var / cov / se: variance / covariance / standard error ρ correlation
$\beta_{1,2}$ slope of X_2 regressed on X_1 σ_ϵ spread around the line